Improving the Trustworthiness of Javascript on the Web

Ezzudin Alkotob* Giulio Berra‡ Benjamin Beurdouche†

Dennis Jackson† Cory Myers‡ Daniel Huigens§

Richard Hansen* Michael Rosenberg¶

*Meta †Mozilla ‡Freedom of the Press Foundation §Proton AG ¶Cloudflare

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Browser E2EE Messaging is Currently Impossible

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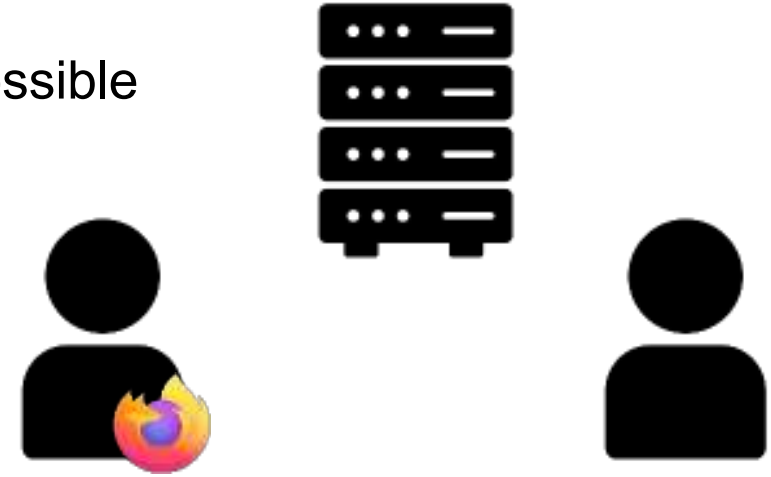

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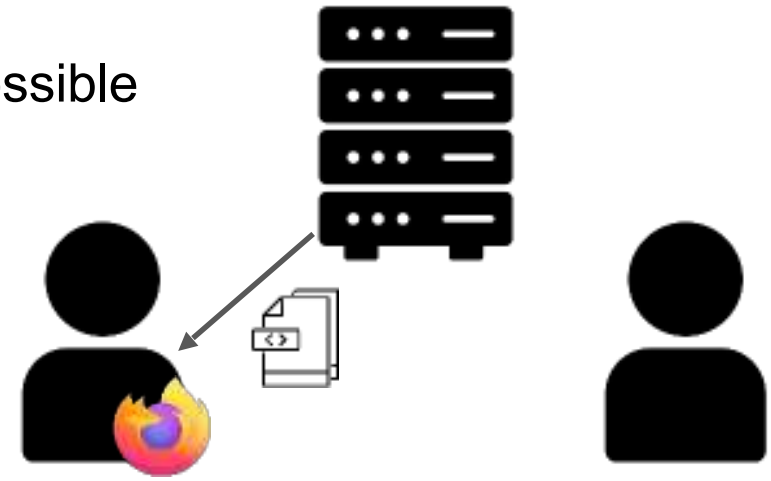

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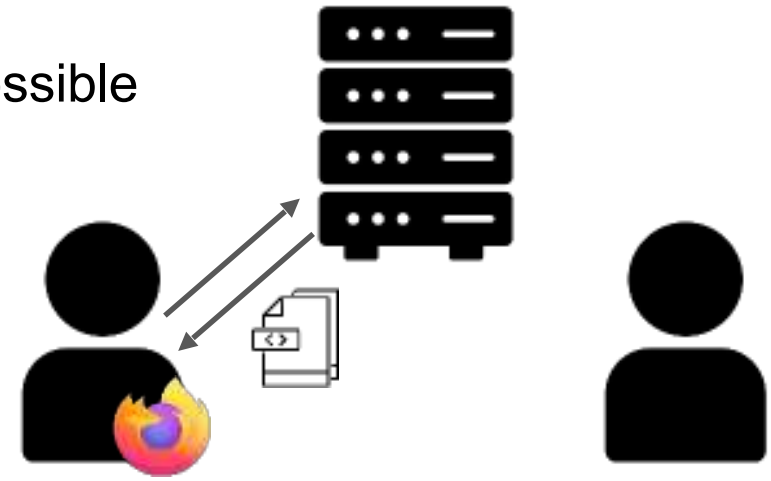

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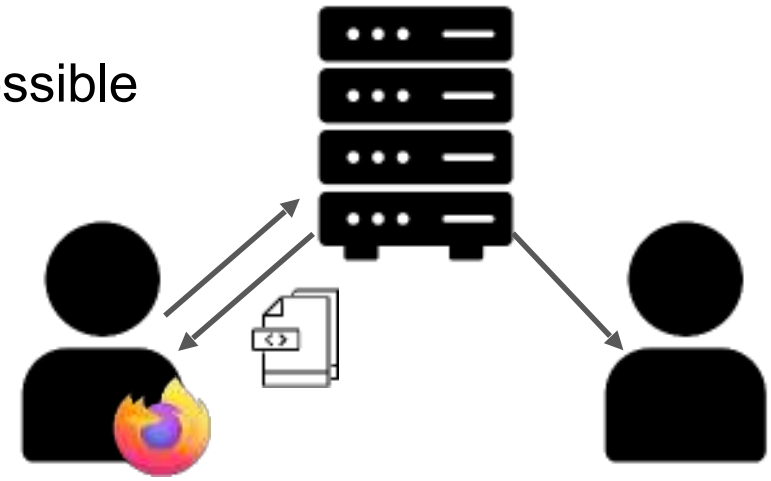

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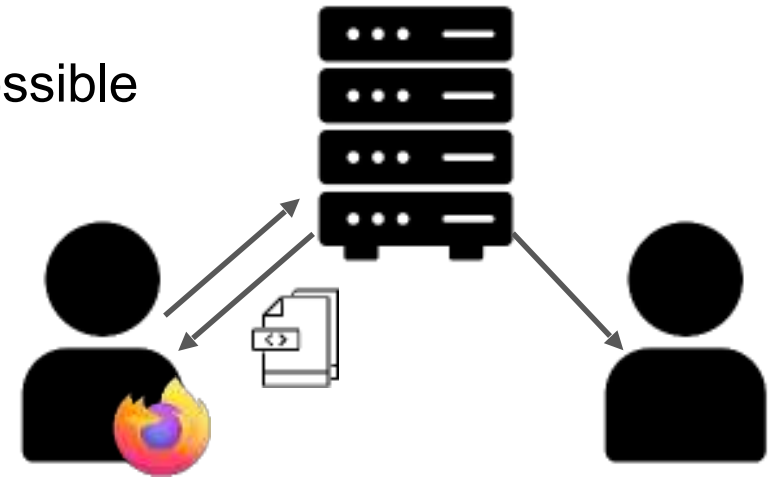

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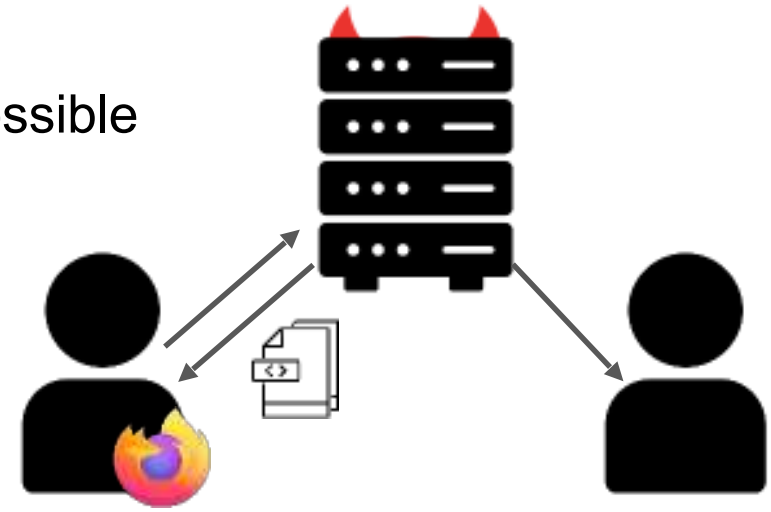

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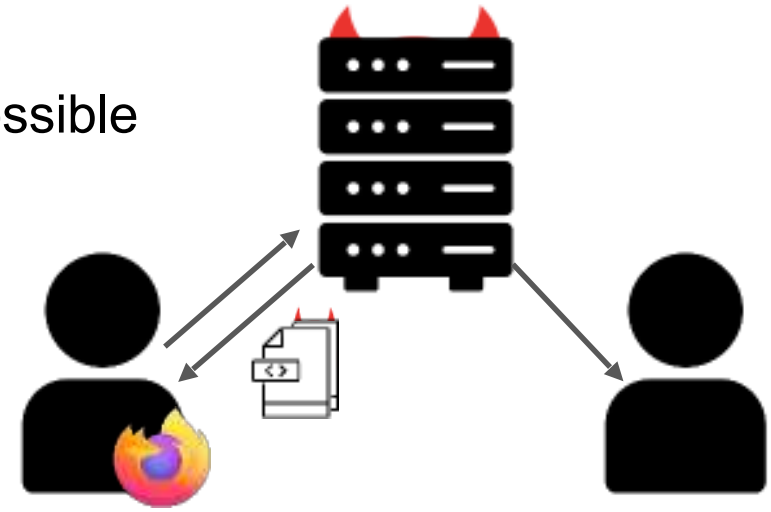

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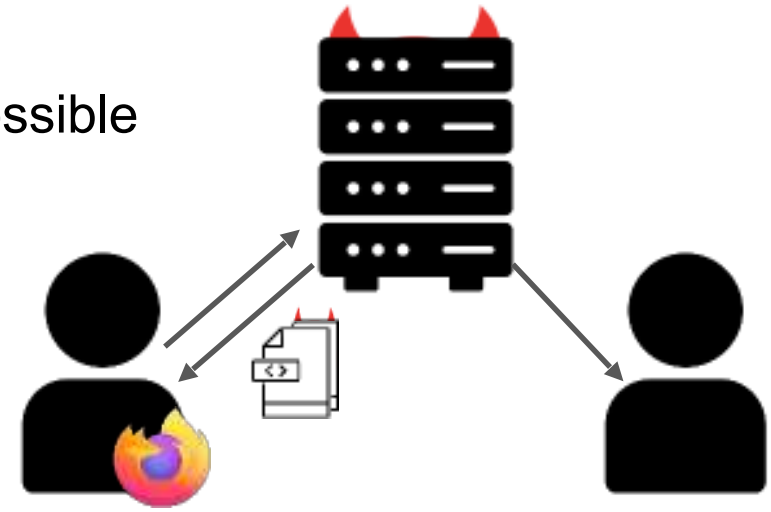

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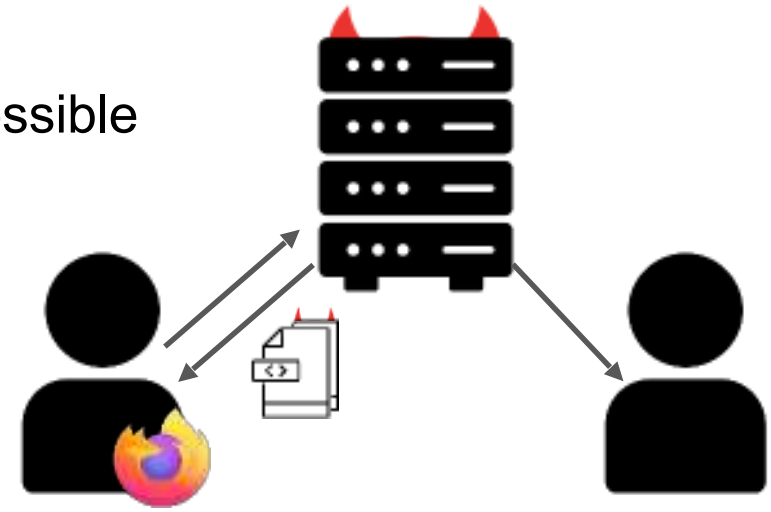

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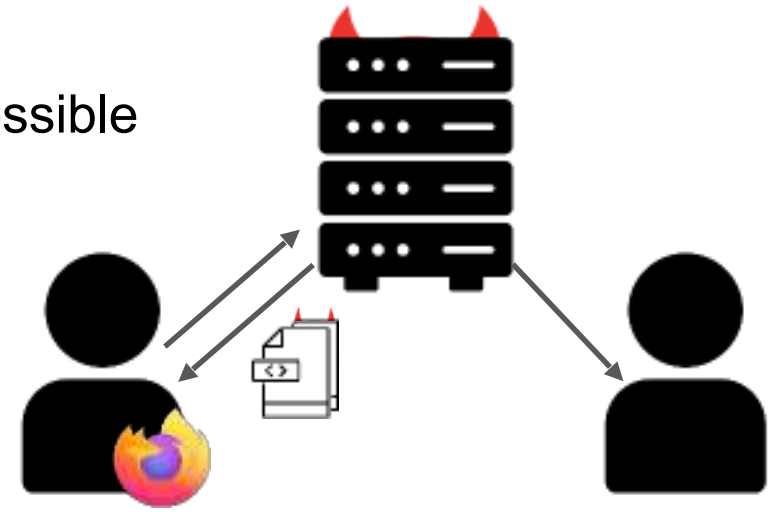

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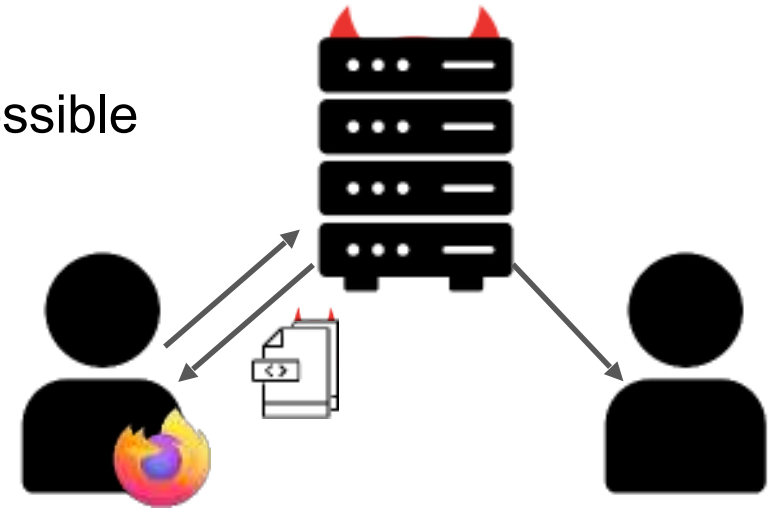

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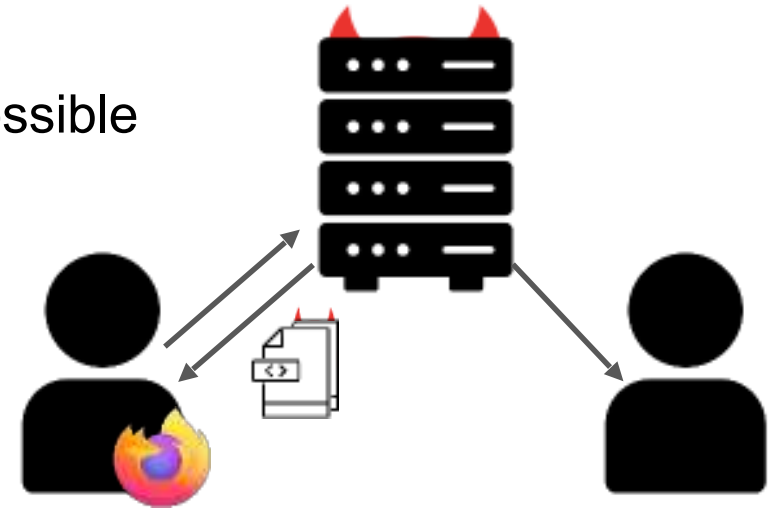

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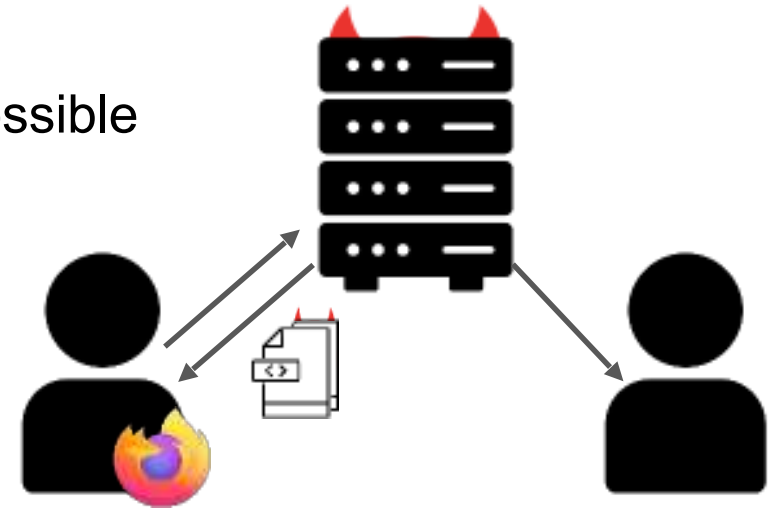

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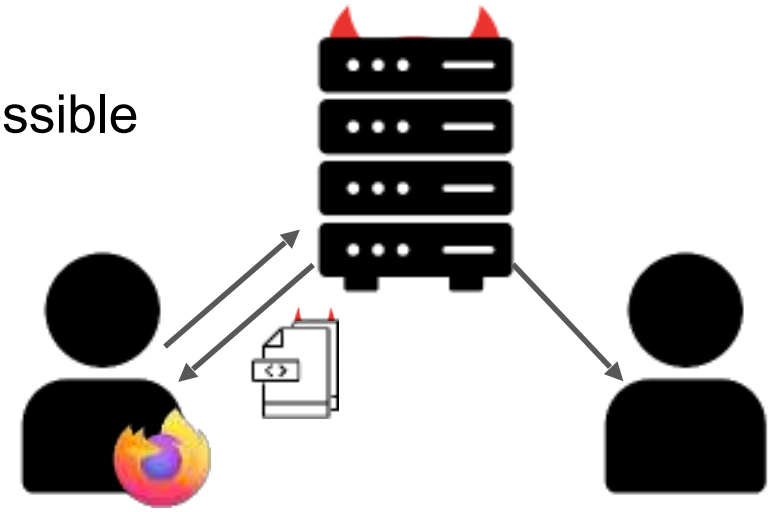

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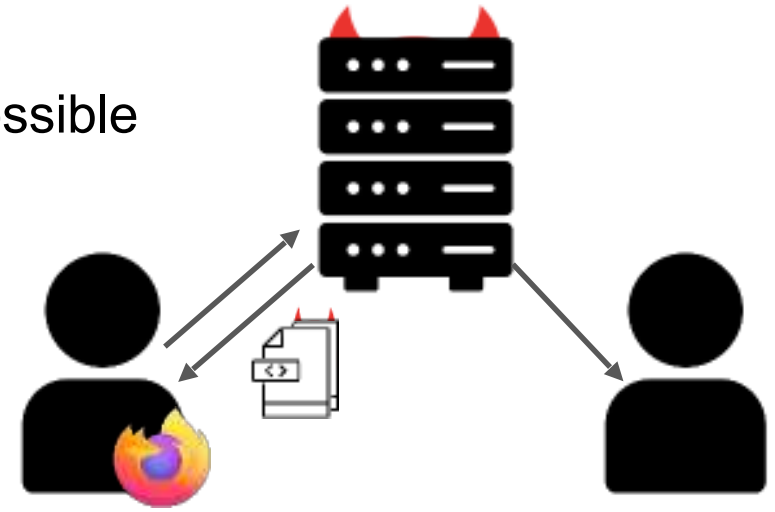

nccgroup

Javascript Cryptography Considered Harmful

If you don't trust the network to deliver a password, or, worse, don't trust the server not to keep user secrets, you can't trust them to deliver security code.

Thomas Ptacek, 2011

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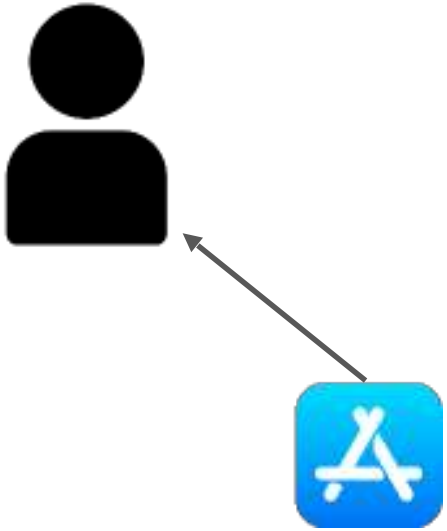

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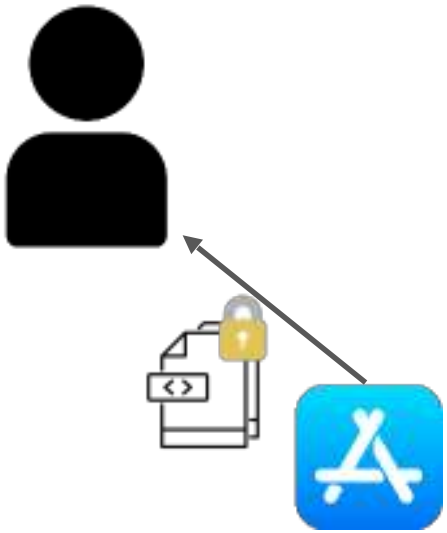

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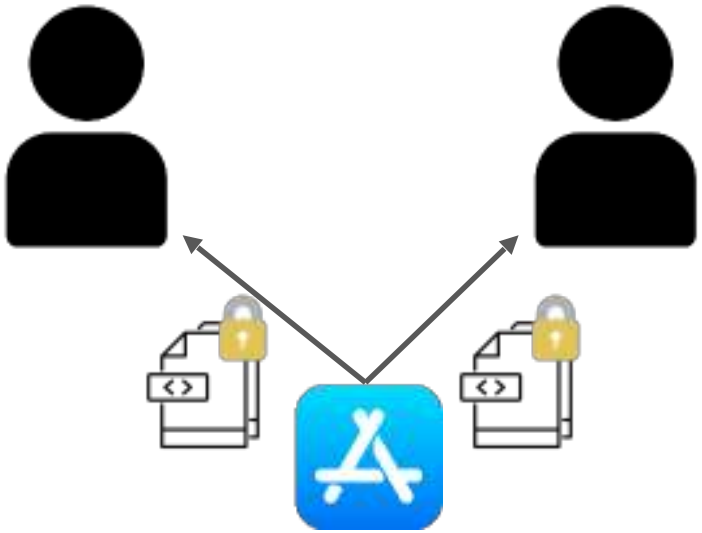

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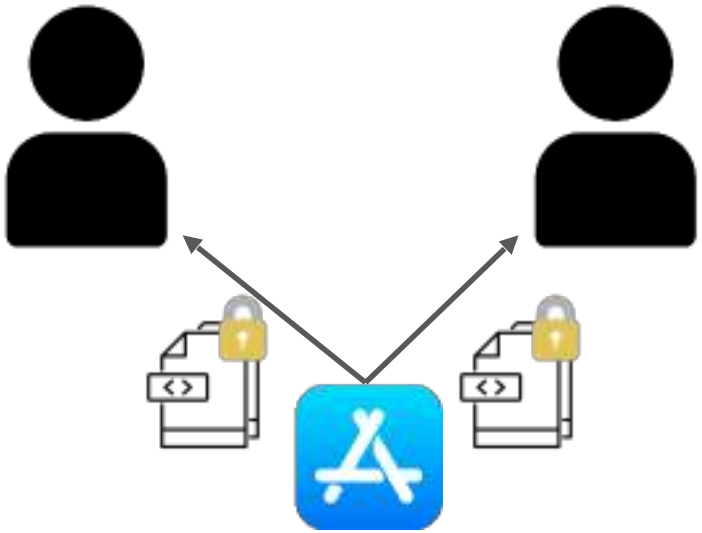

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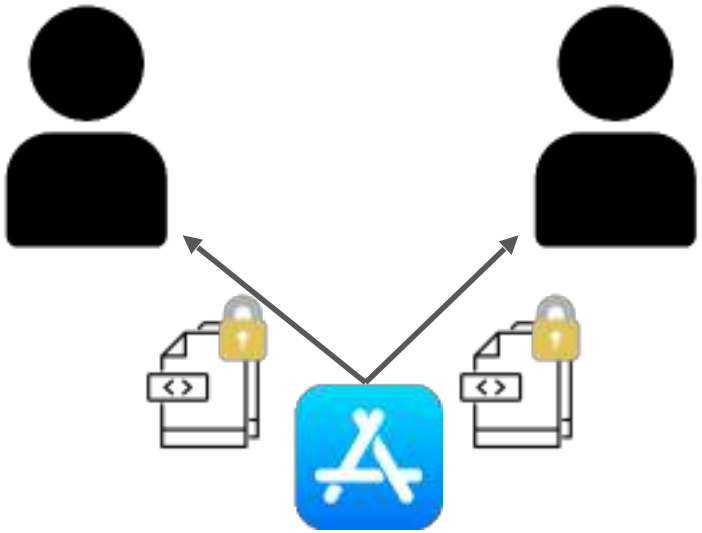

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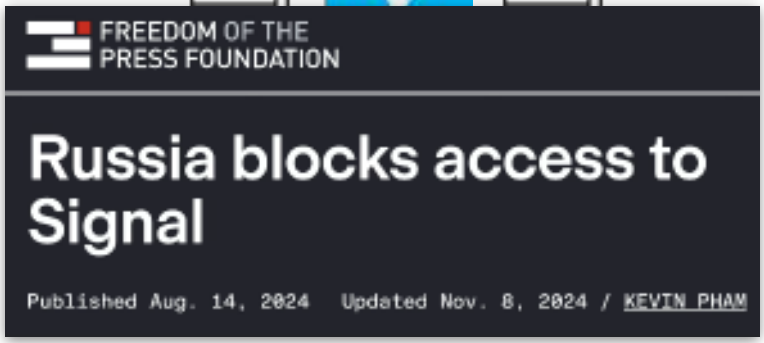

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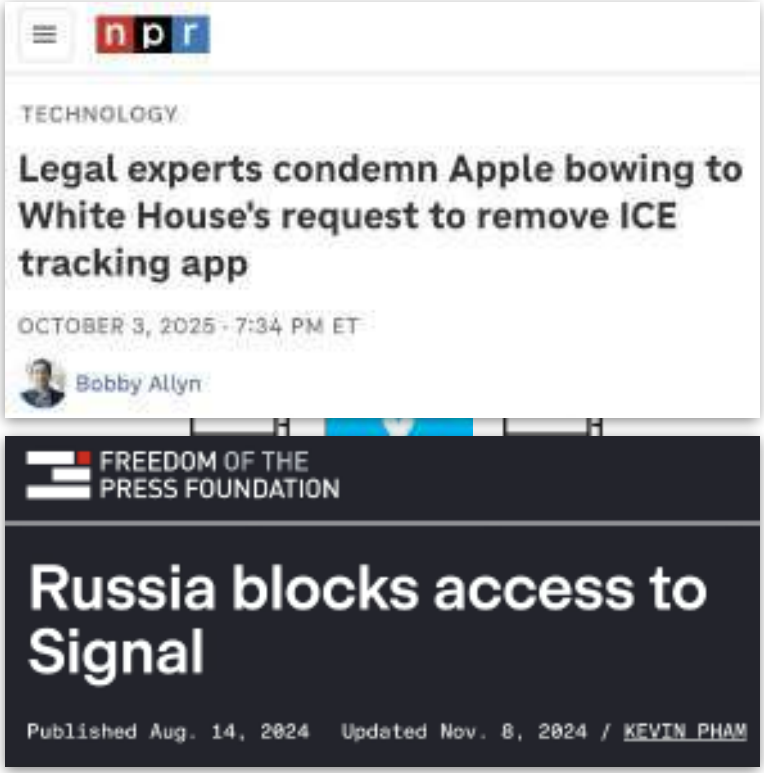

Overview

Bringing Integrity, Consistency, and Transparency to the Web

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WAICT (in progress)

Standardized code transparency for any site

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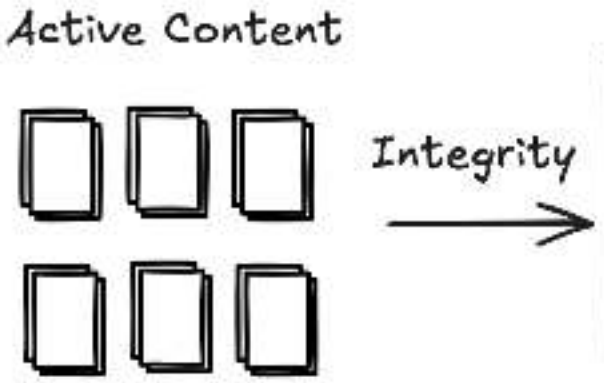

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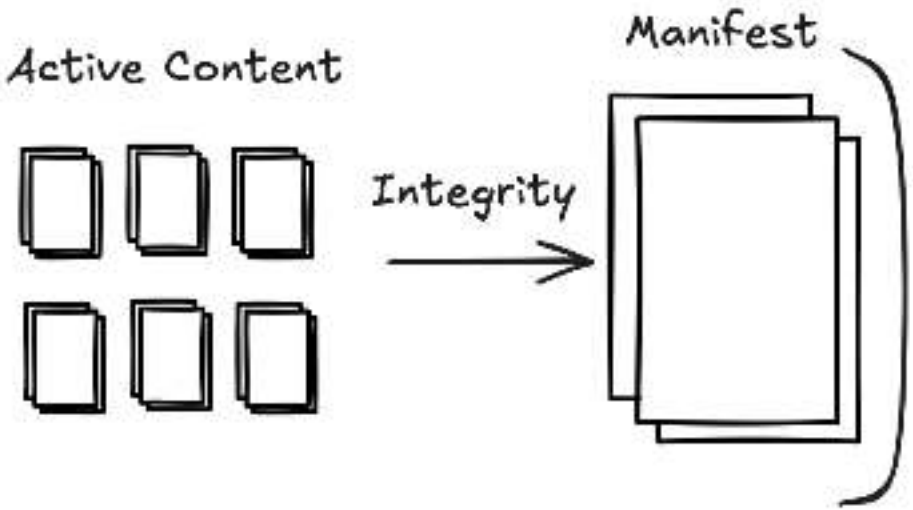

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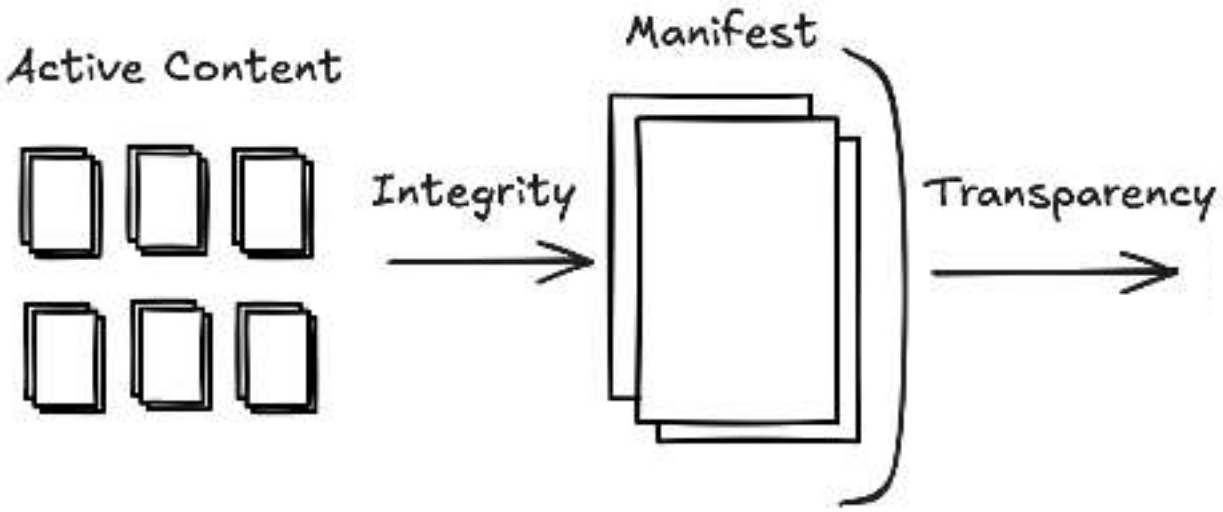

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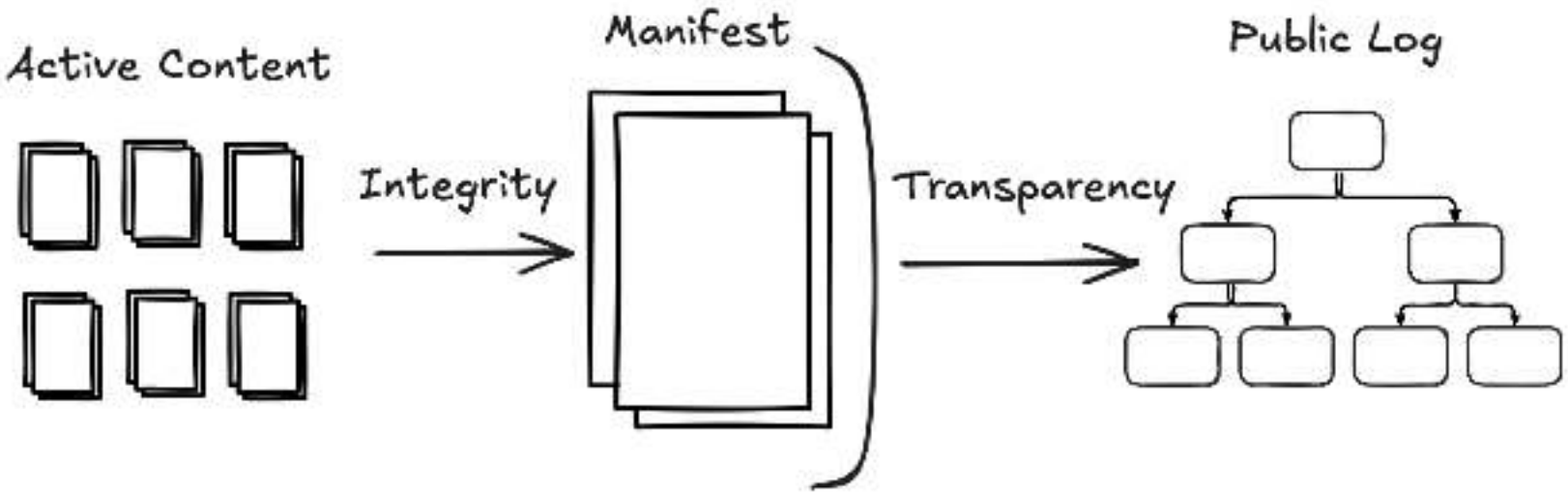

Code Verify

Firefox Browser ADD-ONS

chrome web store

How Code Verify Works

Three-Party Trust Architecture

1. **Meta** publishes web app code + JSON manifest of SHA-256 hashes for every JS/CSS file

2. **Cloudflare** (independent auditor) receives manifest root hash from Meta

3. **Code Verify** extension independently:

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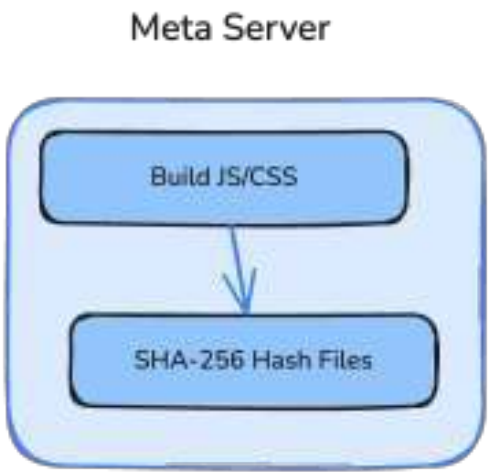

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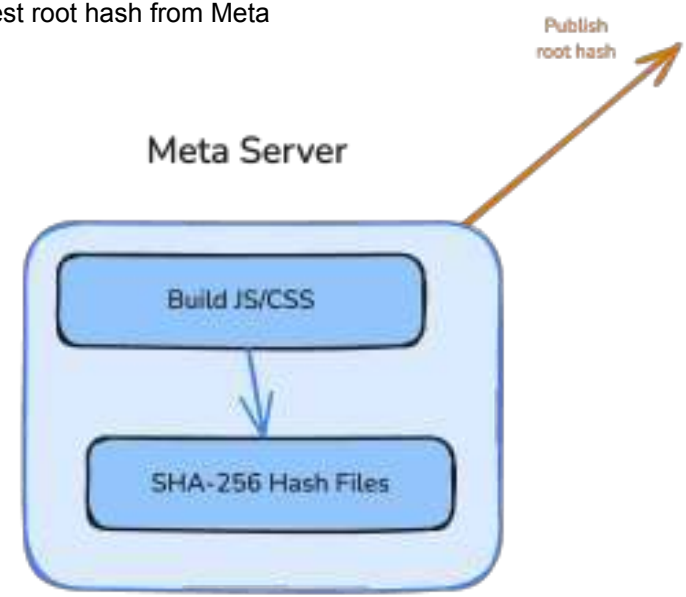

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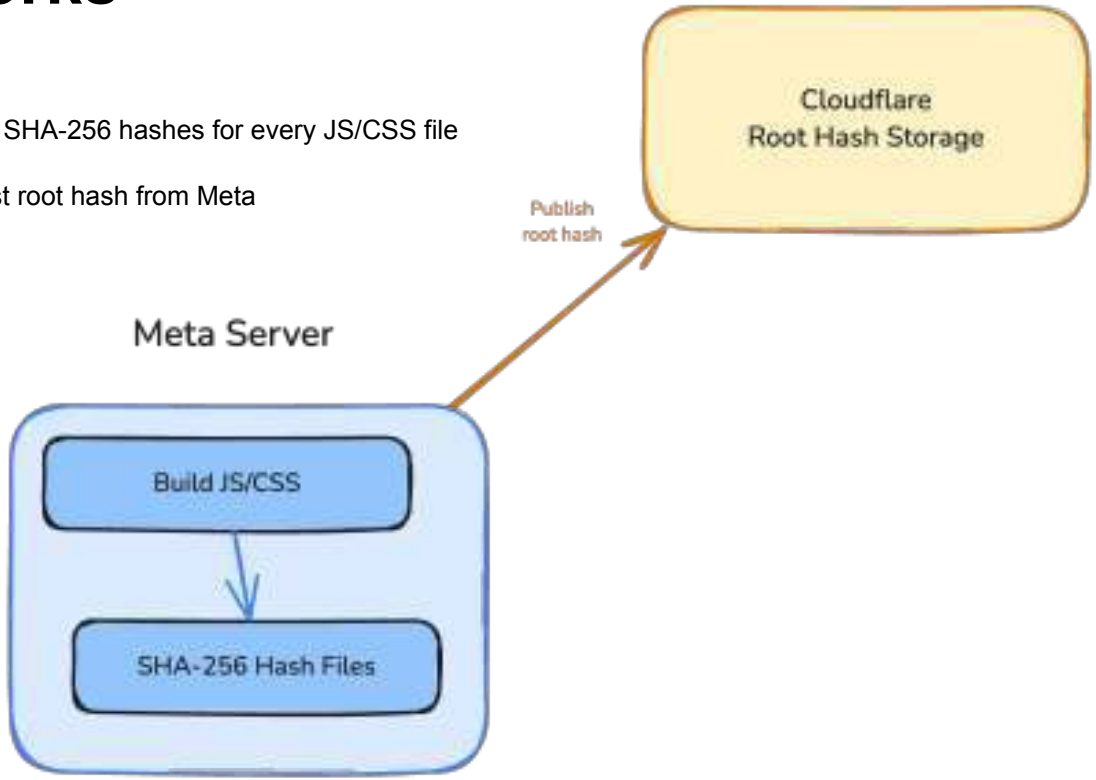

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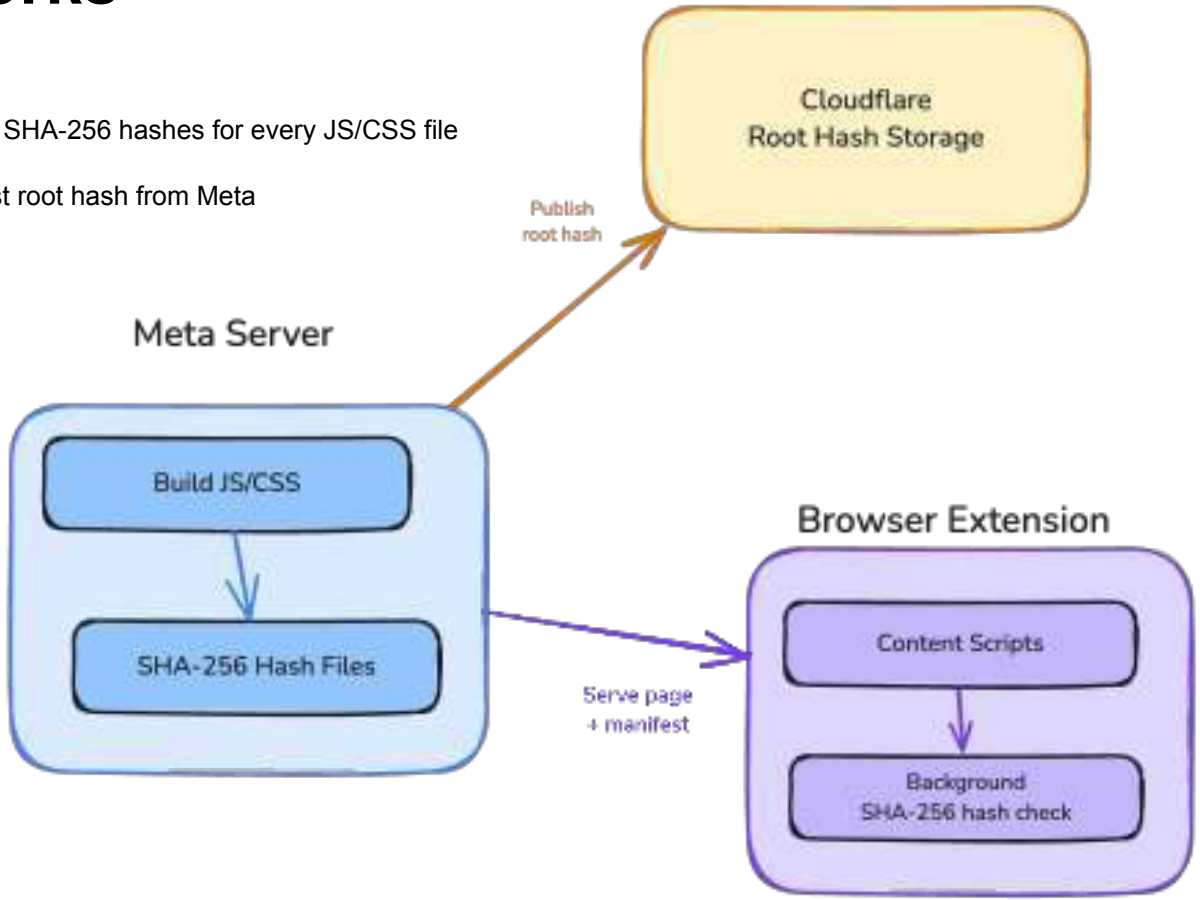

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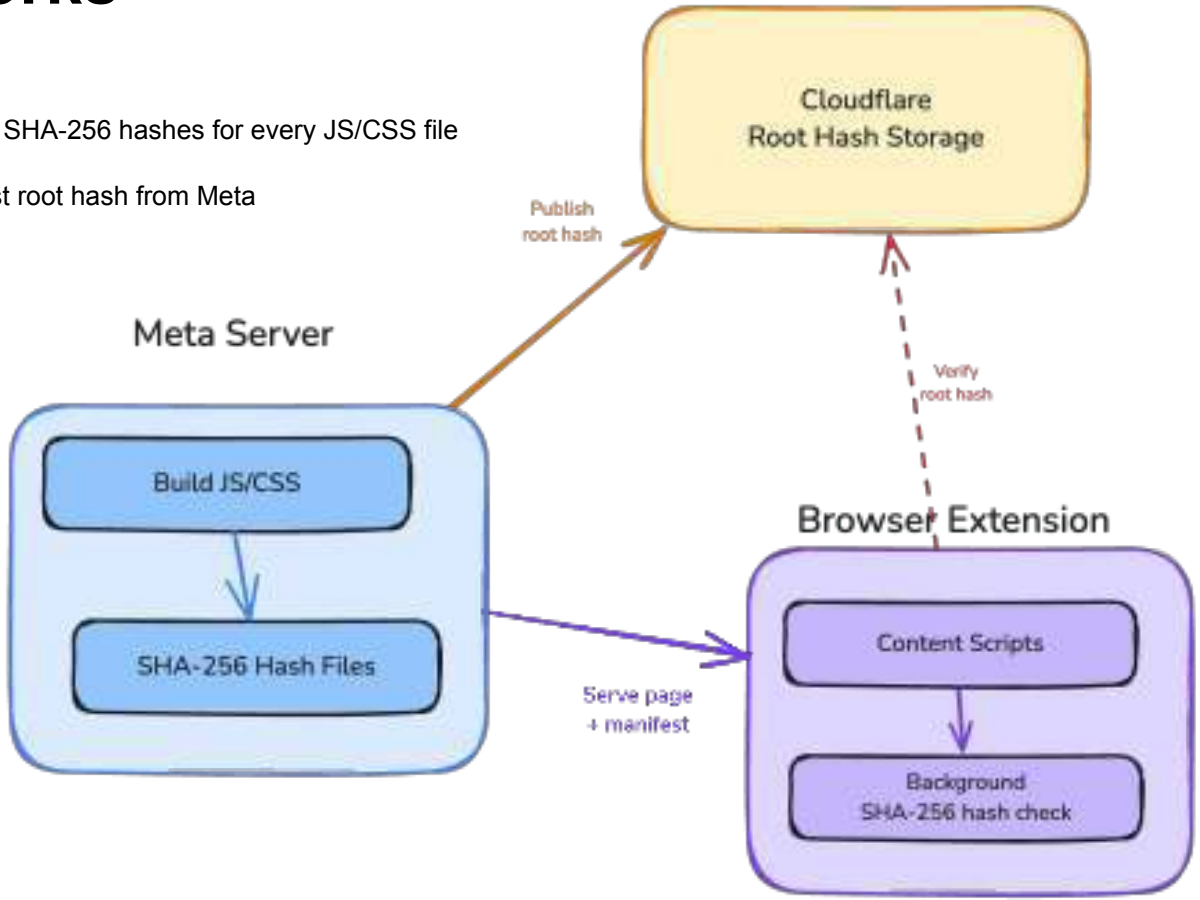

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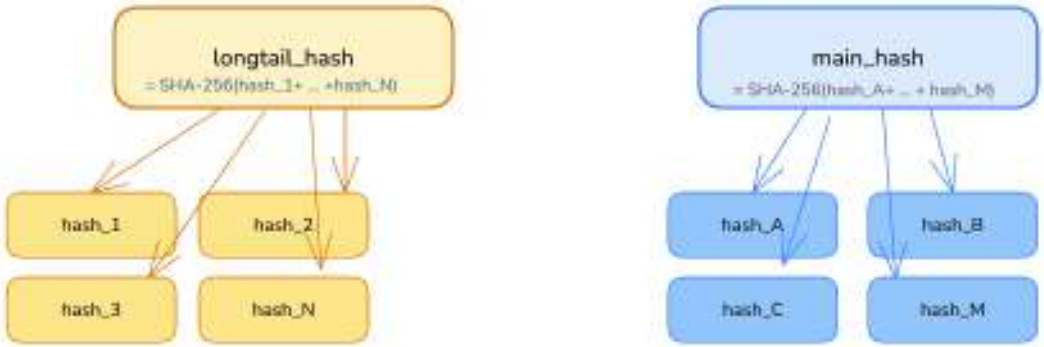

Each leaf = SHA-256 of one JS or CSS file

main = core bundles | longtail = less-frequently-loaded

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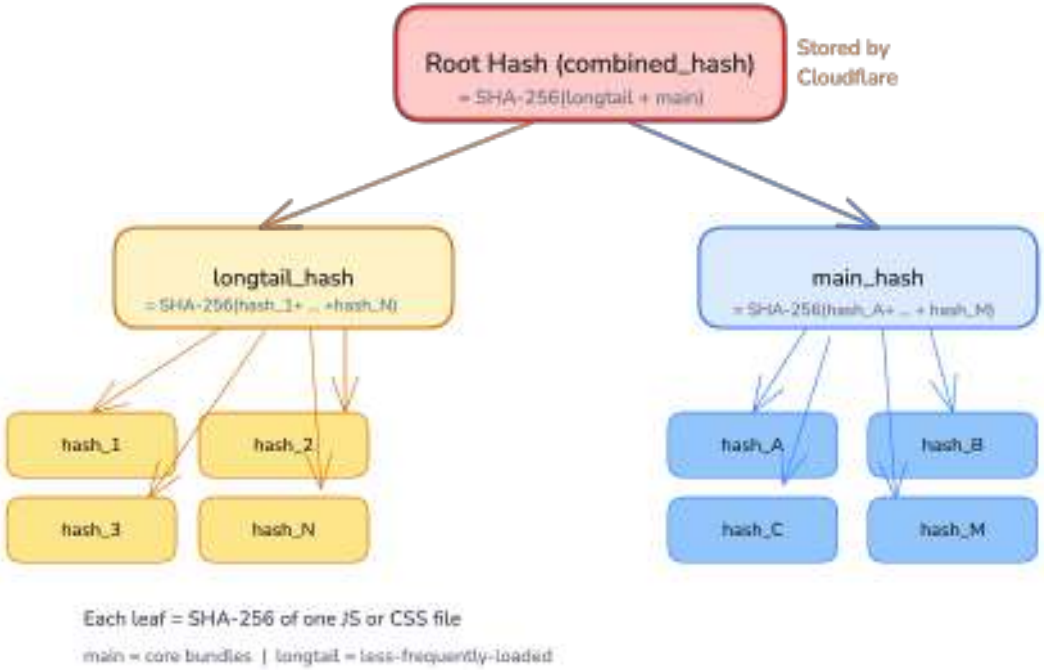

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Step 2: CSP Header Validation

Step 3: DOM Scanning & Monitoring

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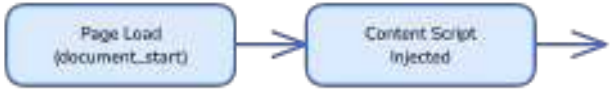

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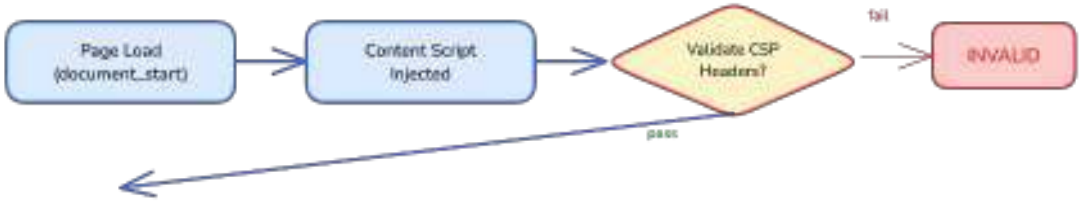

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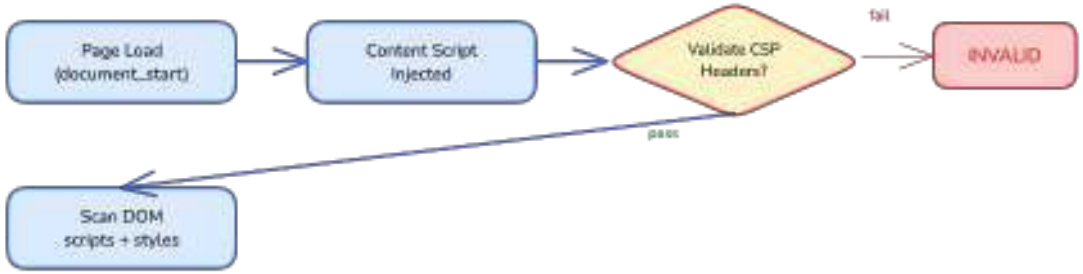

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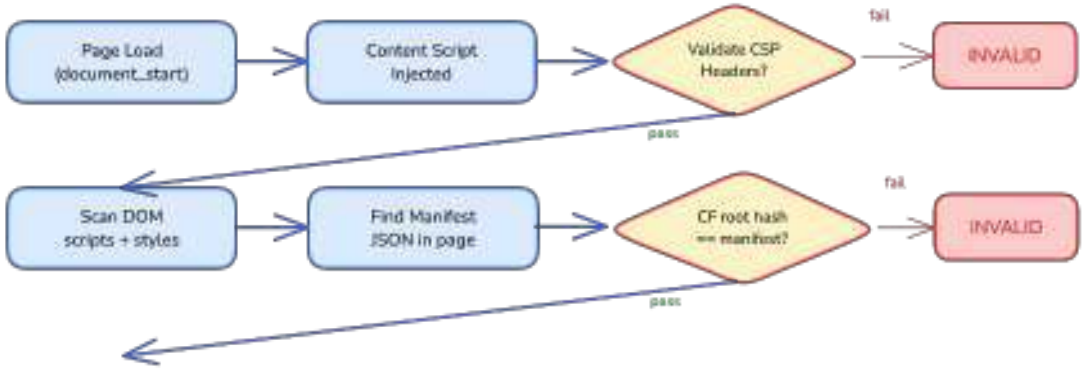

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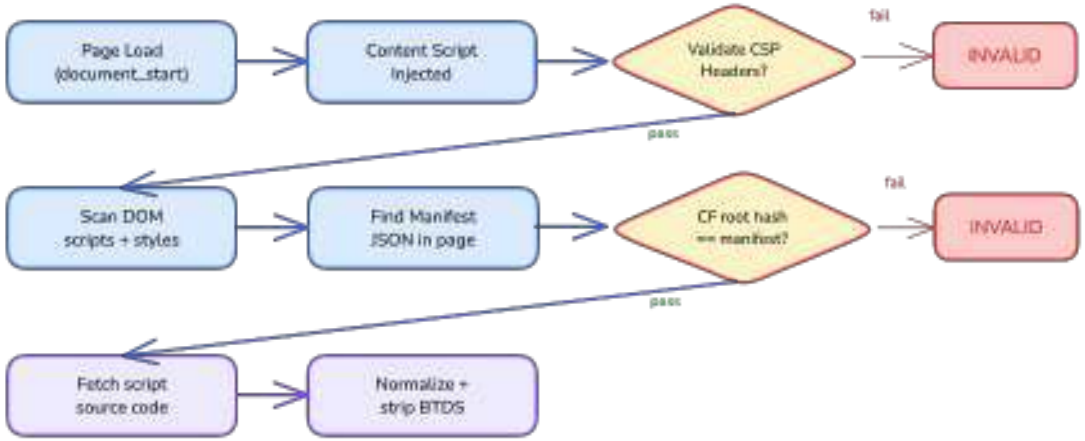

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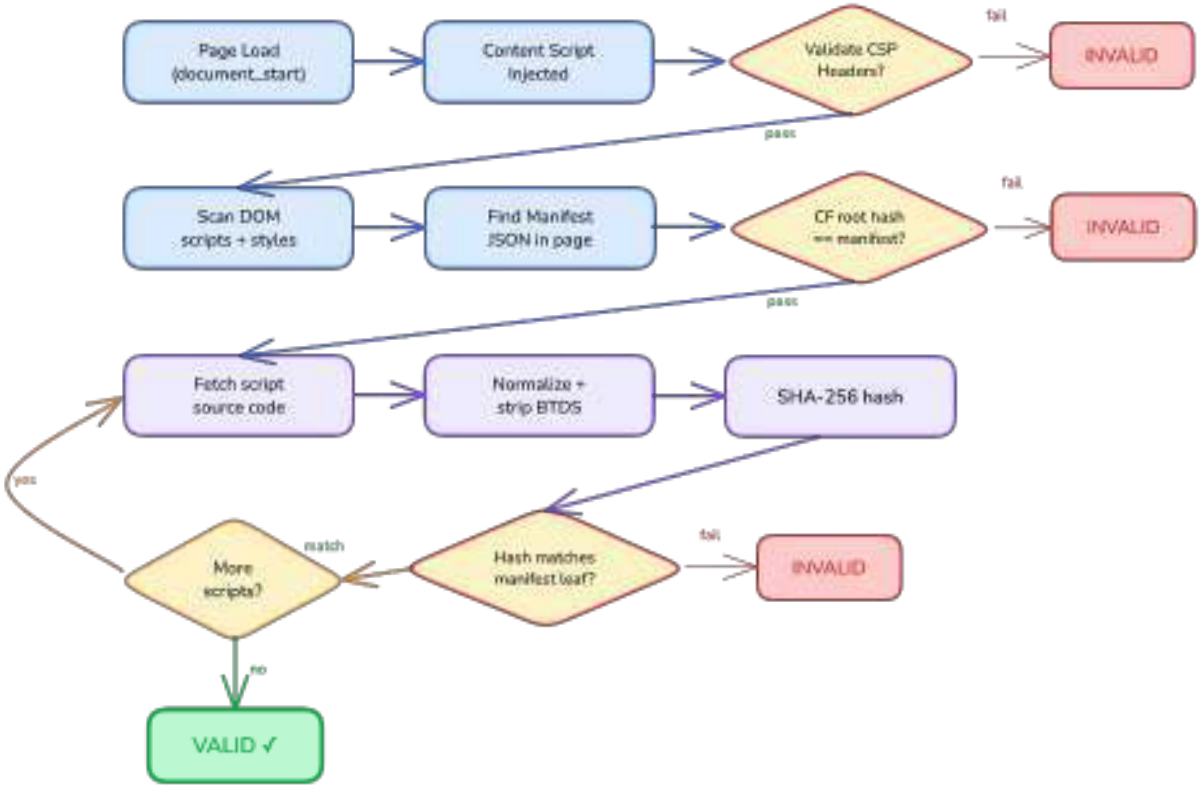
Step 4: Manifest Discovery

Step 5: Cloudflare Verification

Step 6: Fetch & Normalize Scripts/Styles

Step 7: Hashing & Verifying Scripts/Styles



Additional Security Layers

CSP Header Validation

- No unsafe-inline/unsafe-eval in script-src
- Valid worker-src (no blob:, data:, wildcards)

Web Worker CSP and Monitoring

- Detects unverified scripts via importScripts()
- Worker CSP headers independently validated

Cache Integrity Monitoring

- Monitors webRequest responses
- Re-fetch not from cache → tab invalidated
- Prevents TOCTOU/MITM serving different content

WEBCAT

Firefox Browser
ADD-ONS

Let’s talk requirements

1. General-purpose

Let’s talk requirements

1. General-purpose
2. Fail-closed by default

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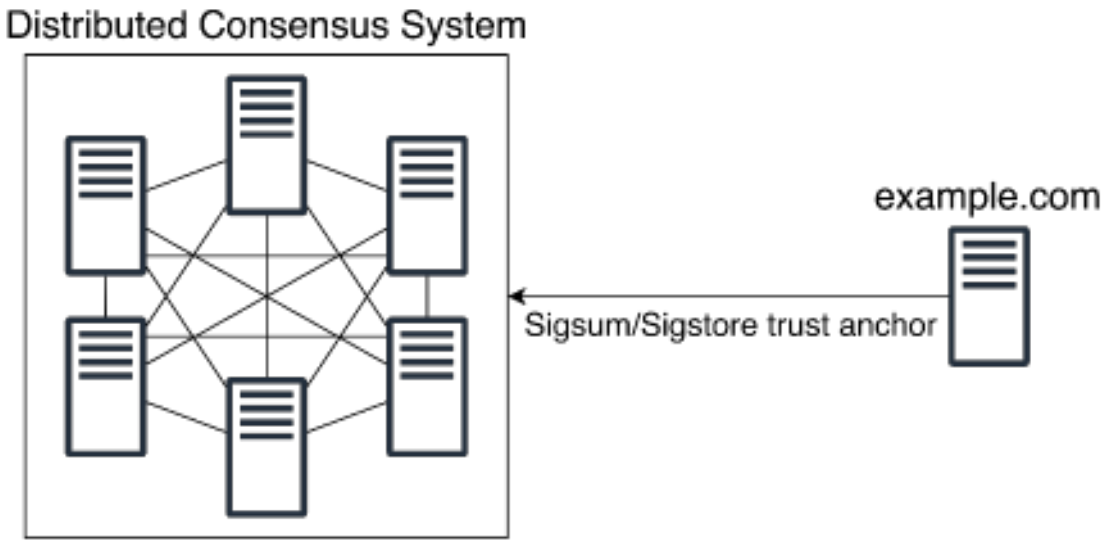
*Focused on **prevention of execution**, detection as a fallback*

Architecture



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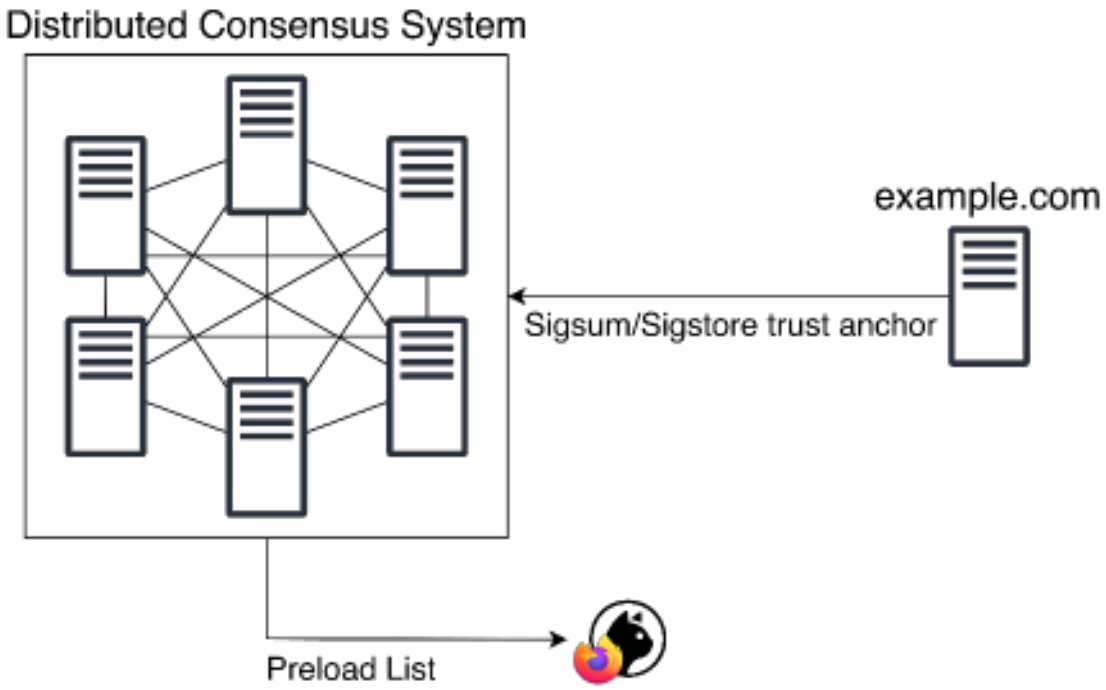

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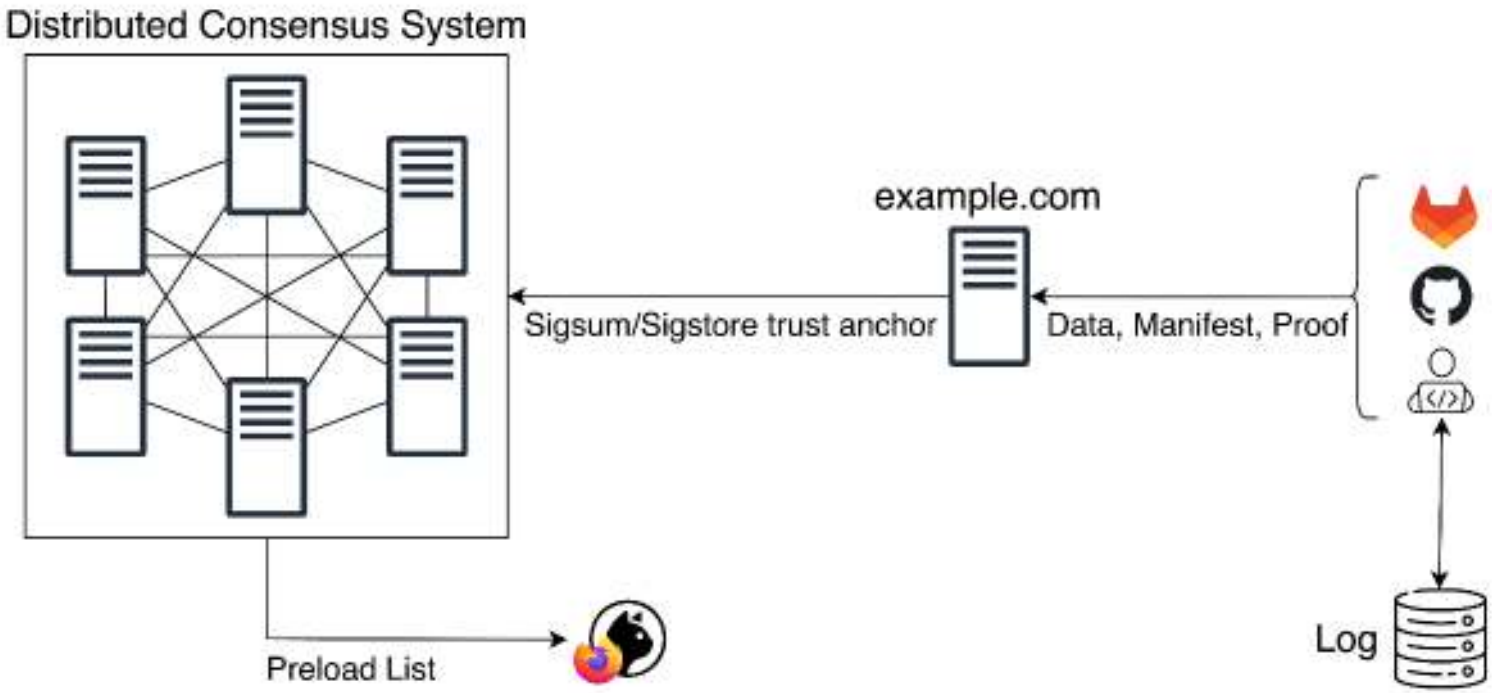
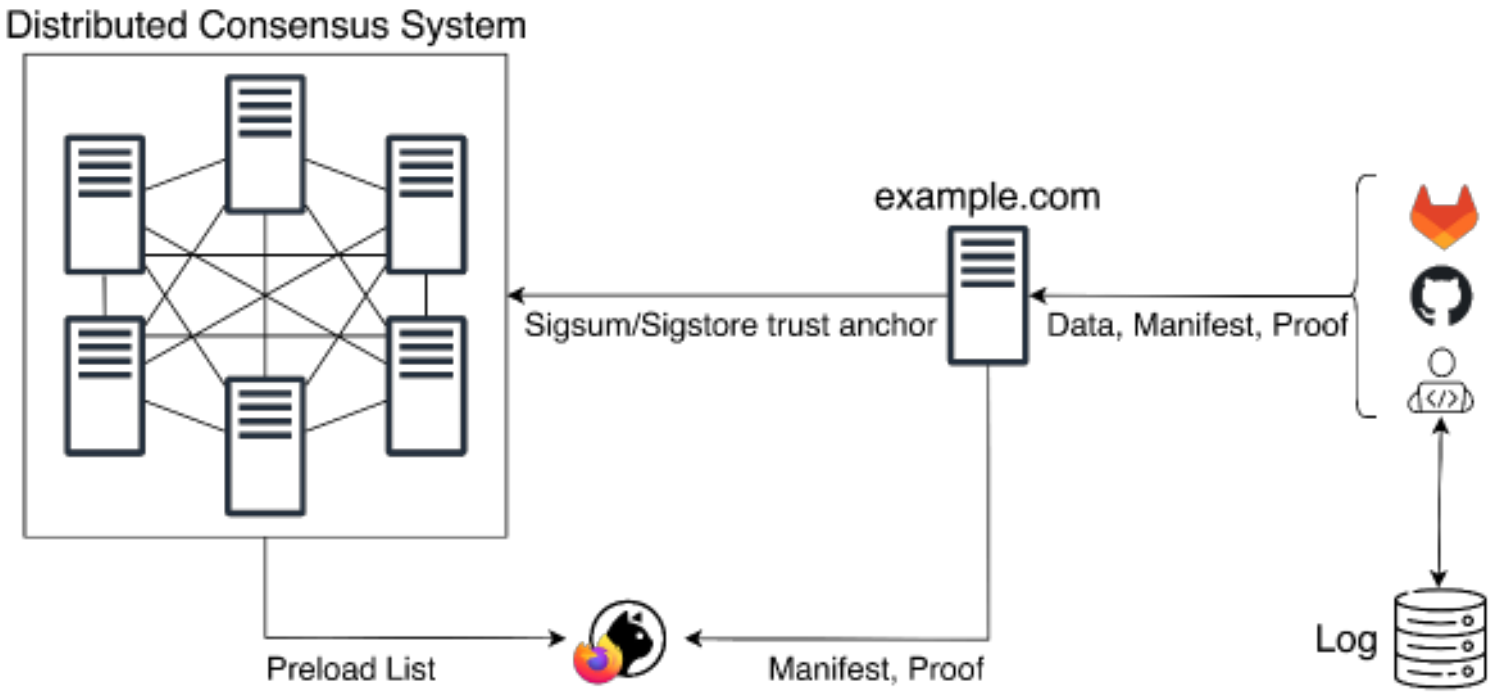
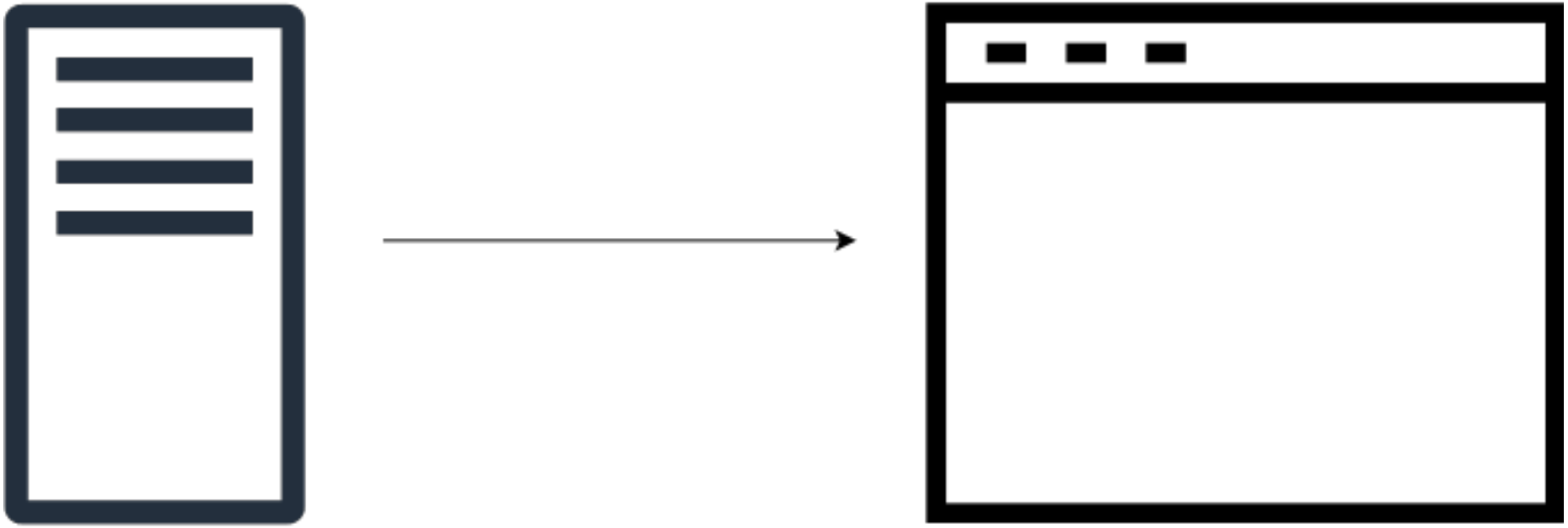


Architecture



What is a Web Application anyway?



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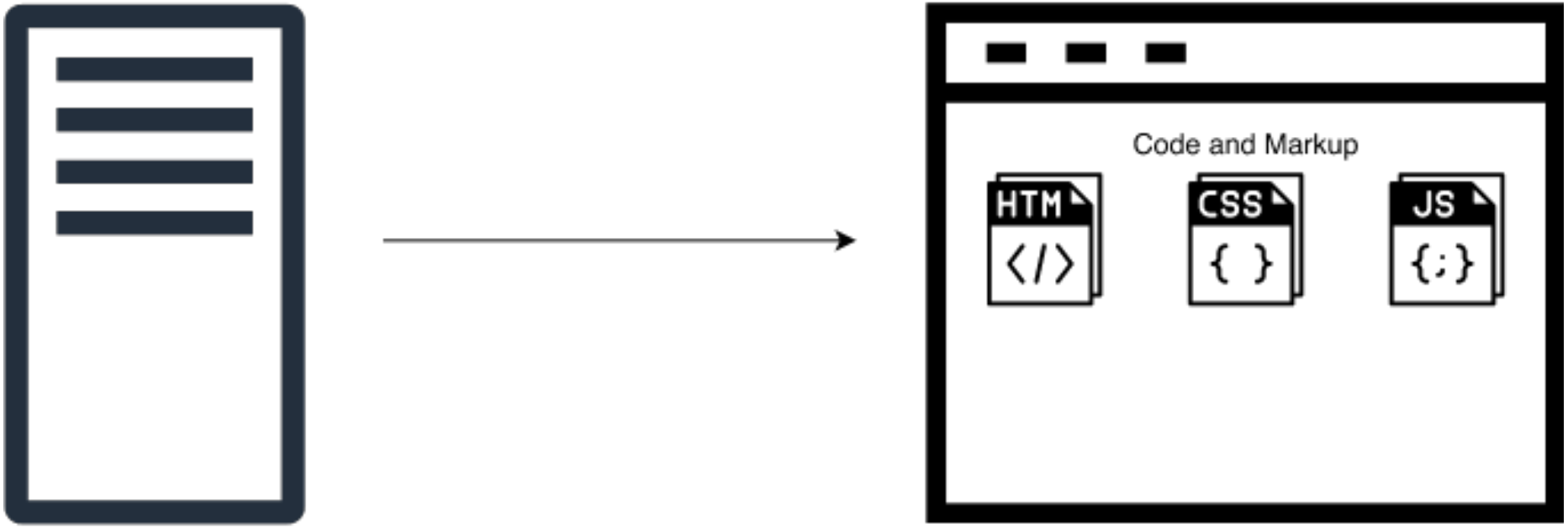

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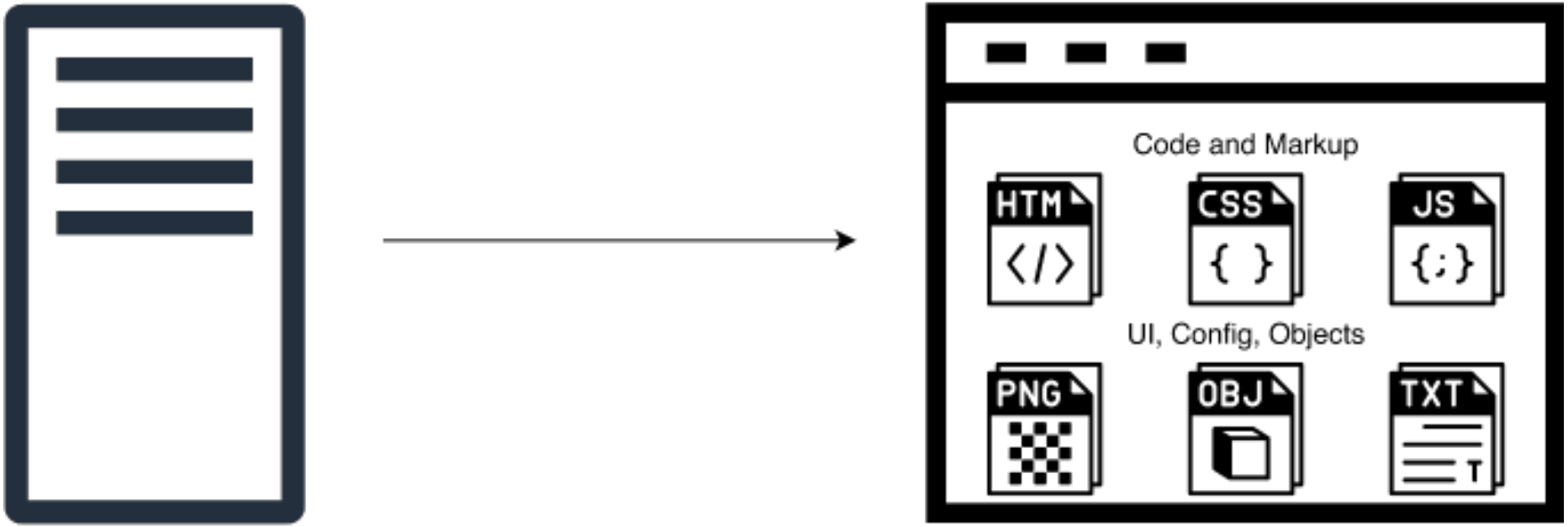
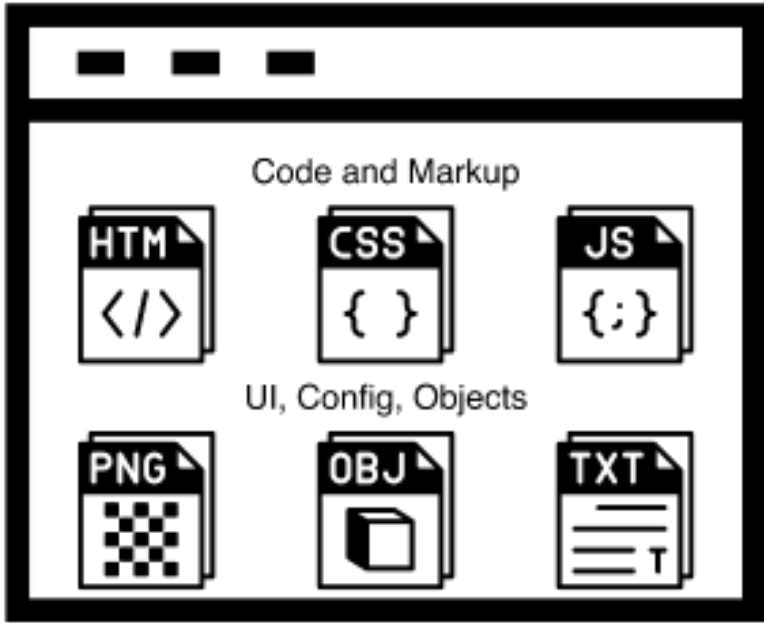


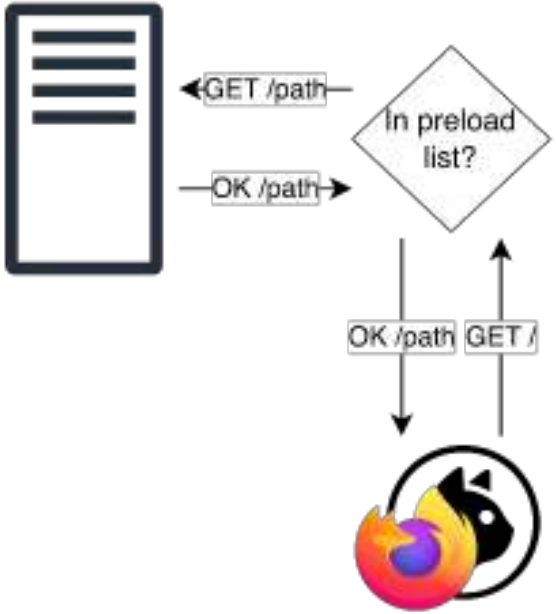
What is a Web Application anyway?

Execution Environment

- Headers (CSP, Location, Link)
- Error pages
- Rewrites



Verification Logic



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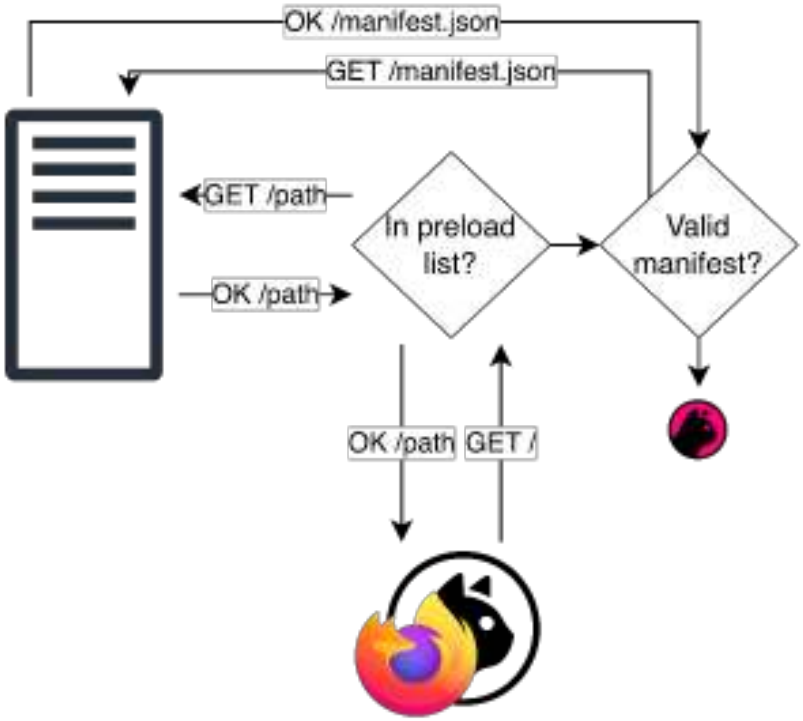

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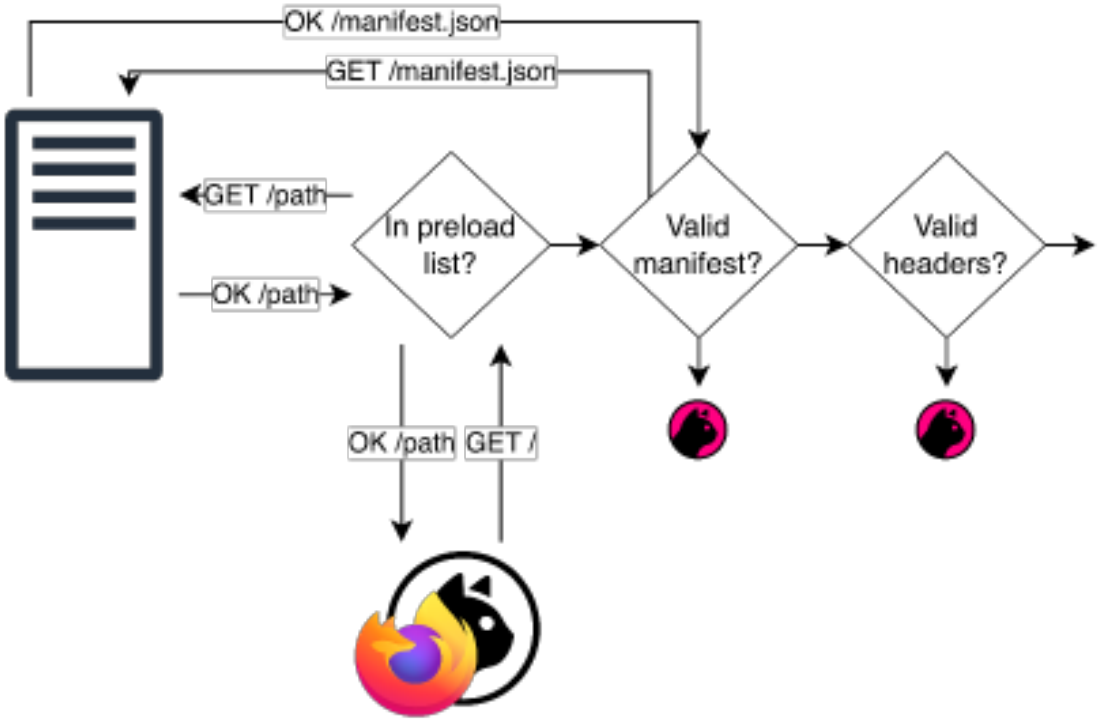

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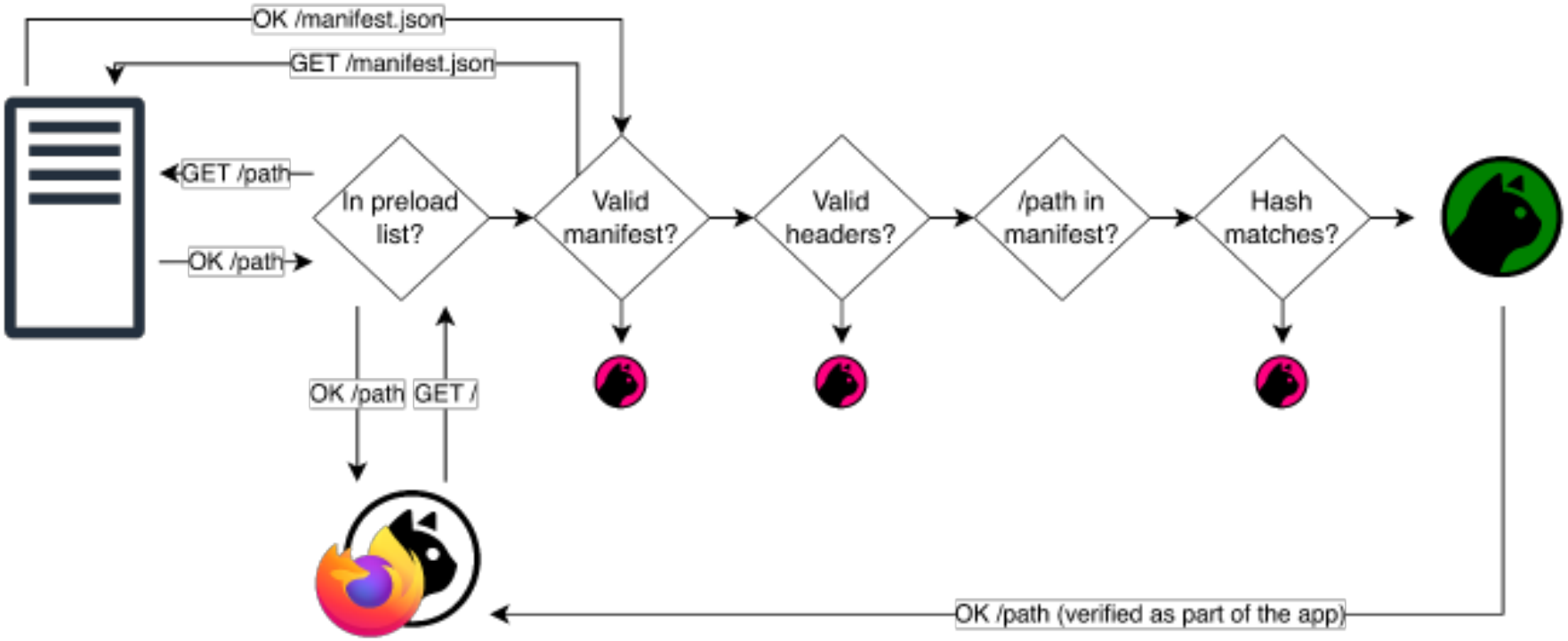

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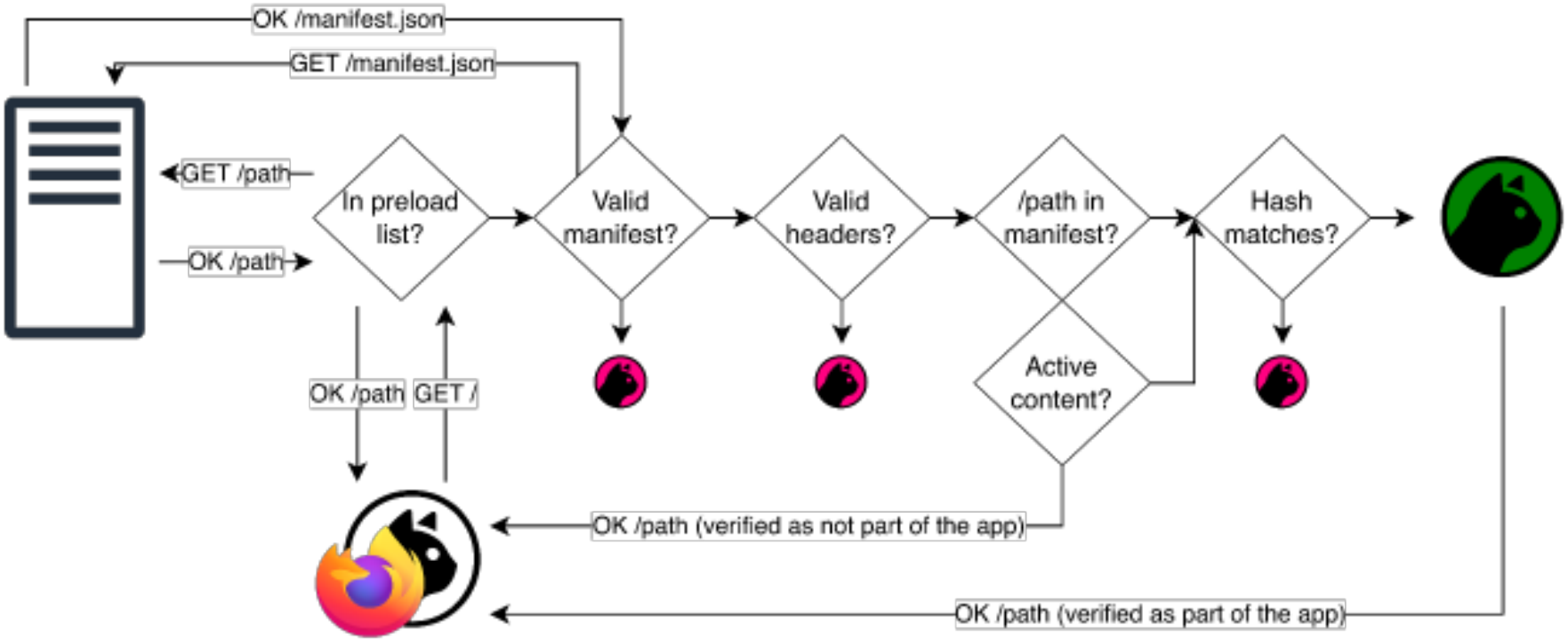

WAICT

Web Application Integrity, Consistency and Transparency

1. Standardize these approaches
2. Integrate into the browser

Deploying WAICT

Opt-in Security Policy via HTTP Header

Deploying WAICT

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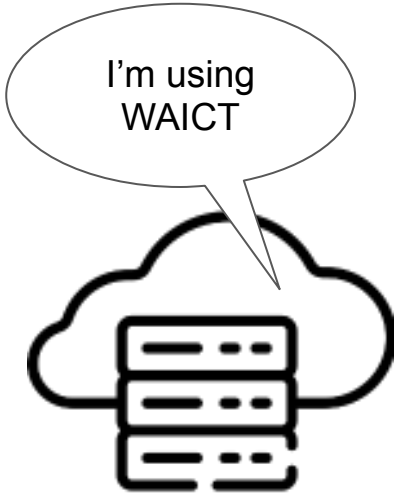
Opt-in Security Policy via HTTP Header

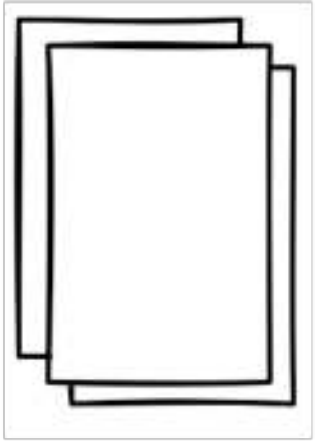
Deploying WAICT




```
Integrity-Policy-WAICT-v1:
    max-age=86400,
    mode=enforce,
    preload=?0,
    manifest="/waict/1.json"
```

Opt-in Security Policy via HTTP Header

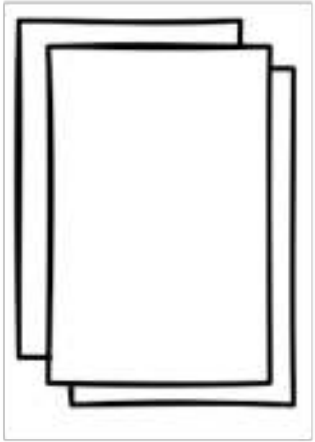
Deploying WAICT



Manifest

No single point of failure

Deploying WAICT



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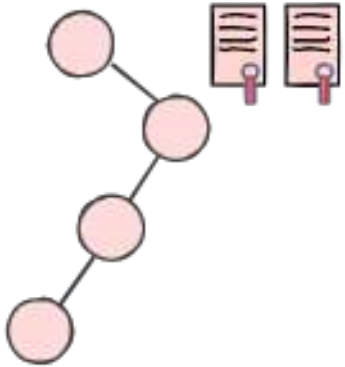
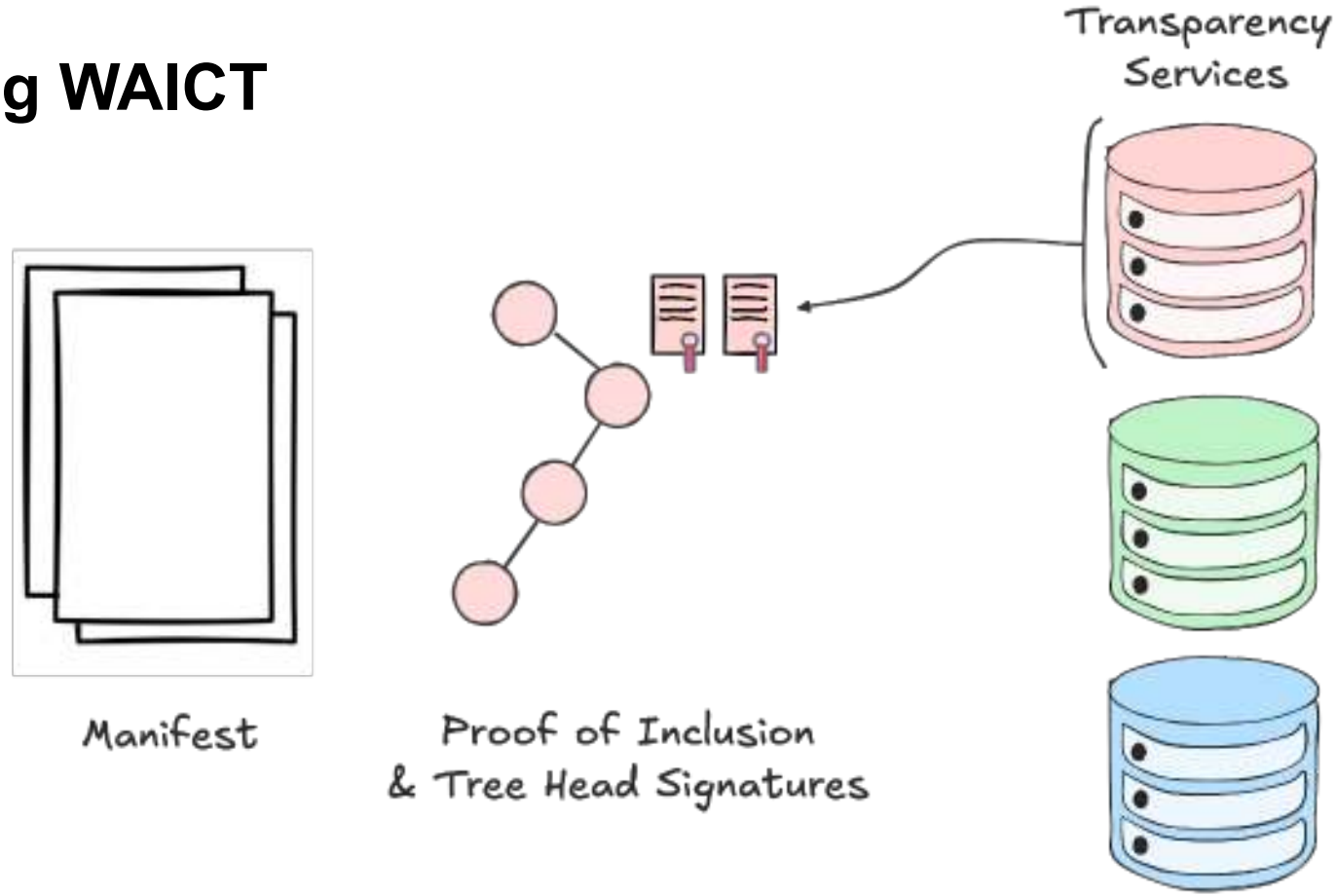


Proof of Inclusion
& Tree Head Signatures

No single point of failure

Deploying WAICT



No single point of failure

Goal: Transparency

Detection, not prevention!

Our Transparency Logs are fancy

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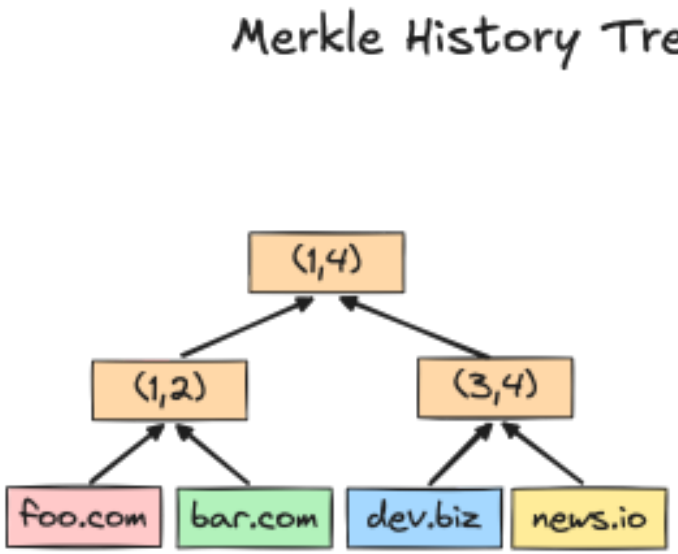

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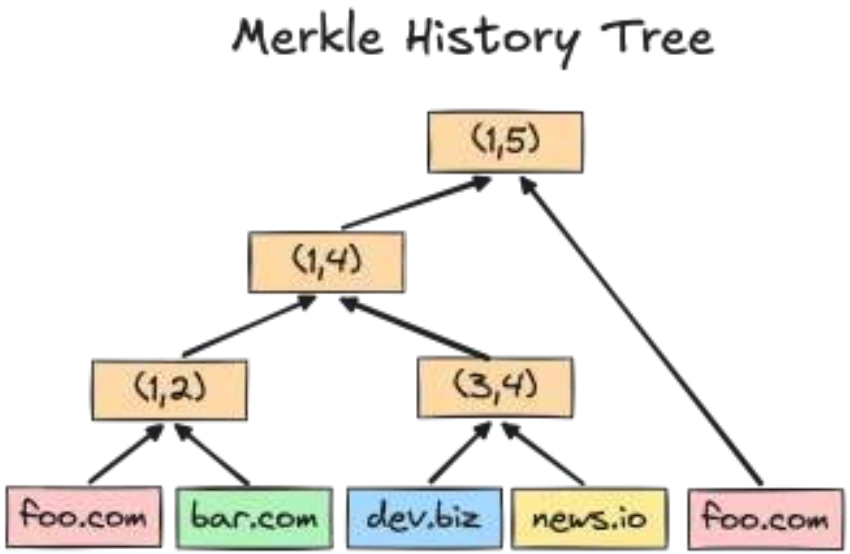

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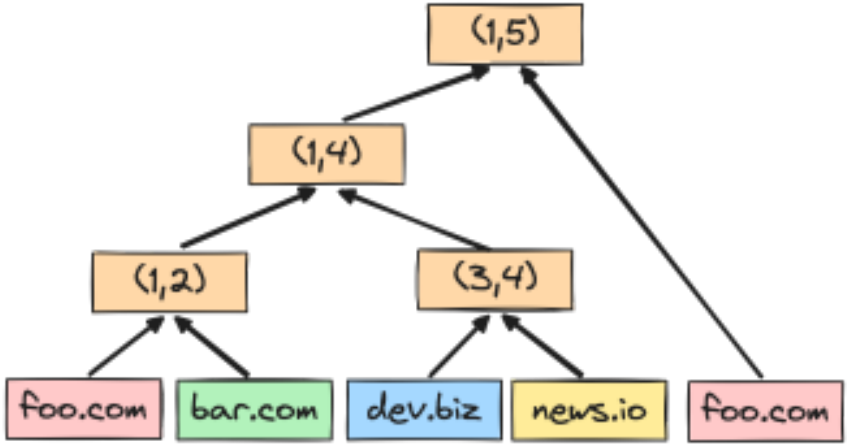

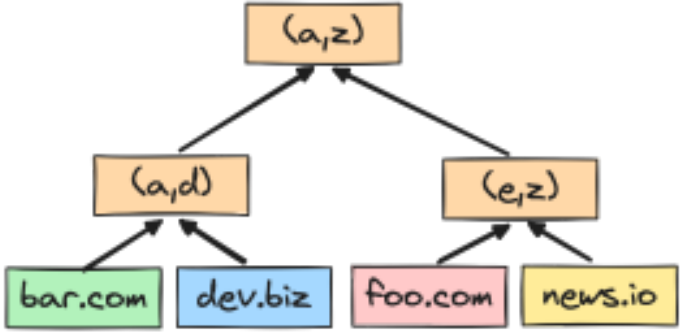

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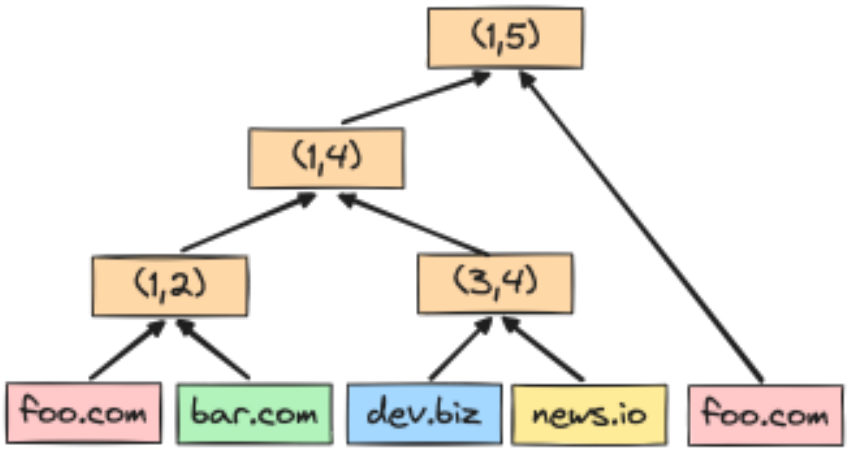

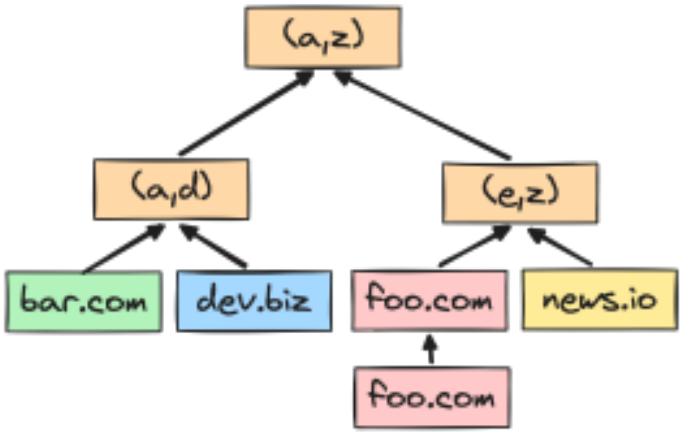

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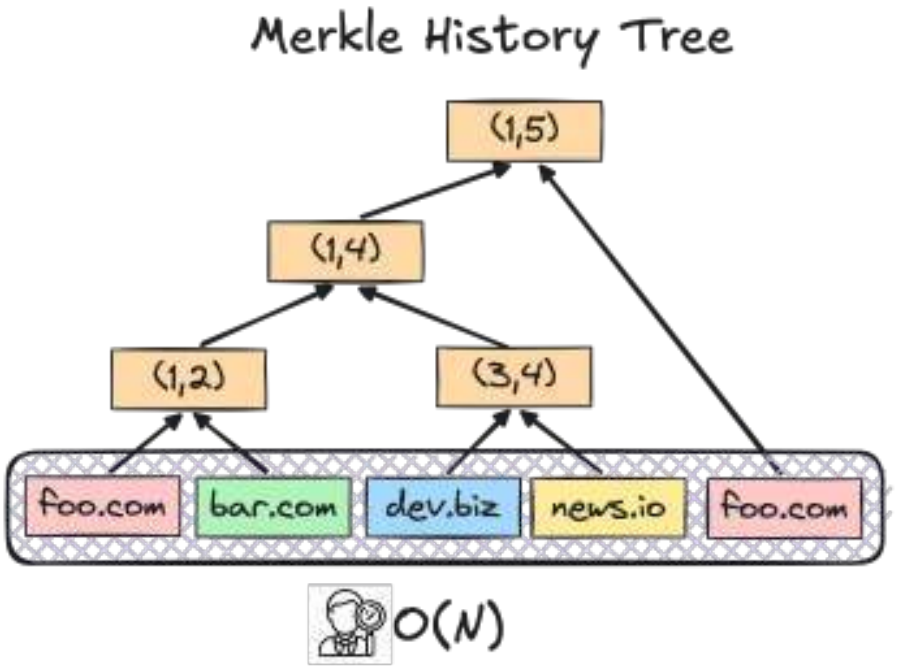

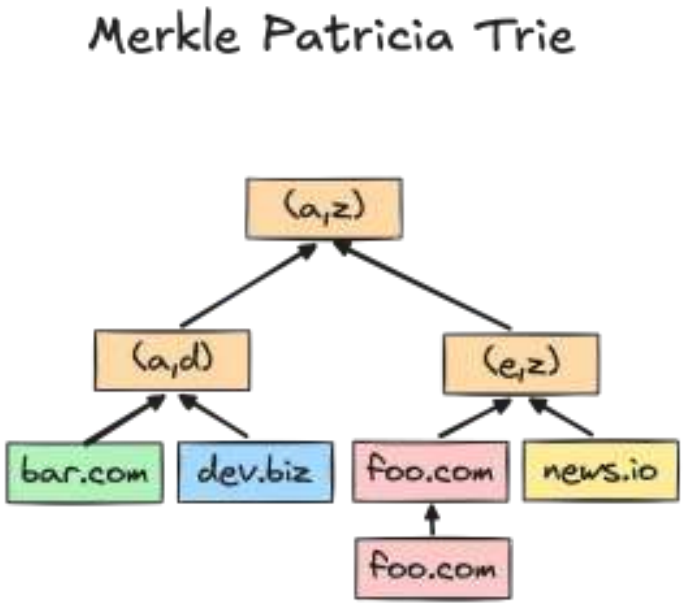

Structured logs enable efficient auditing

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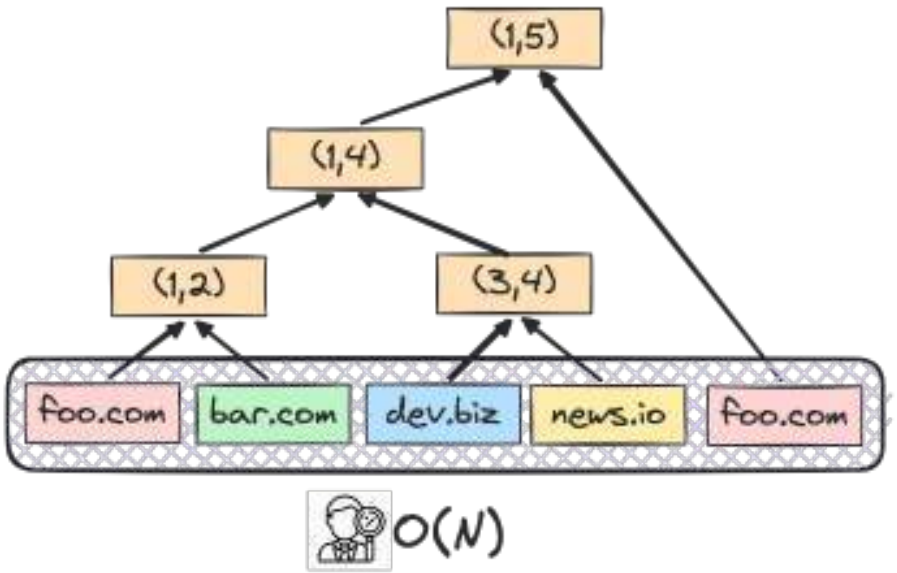

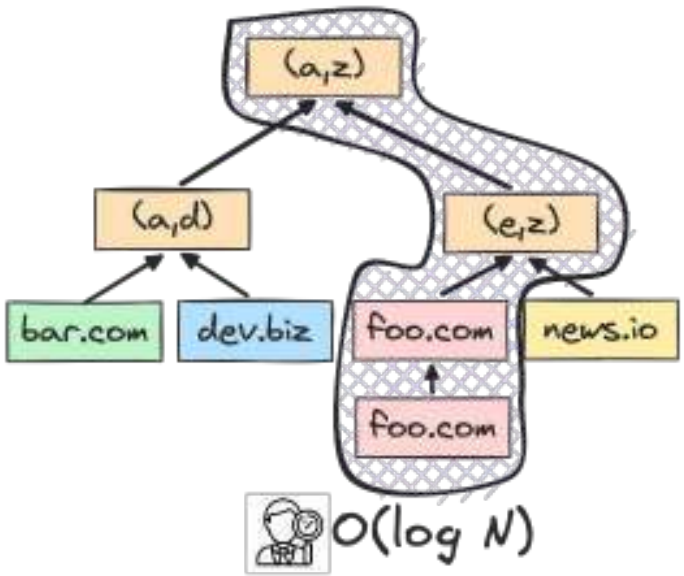

Structured logs enable efficient auditing

Accidents happen

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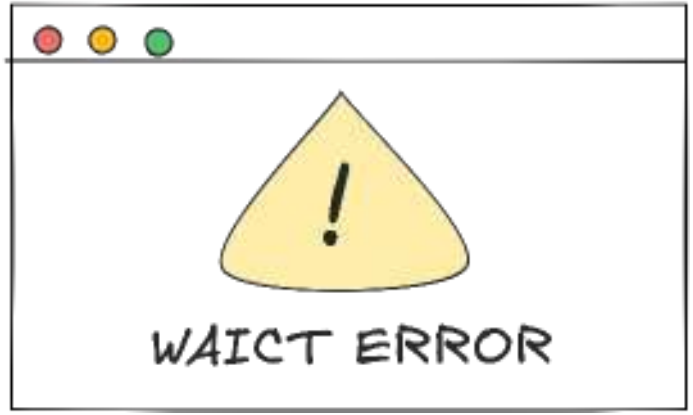

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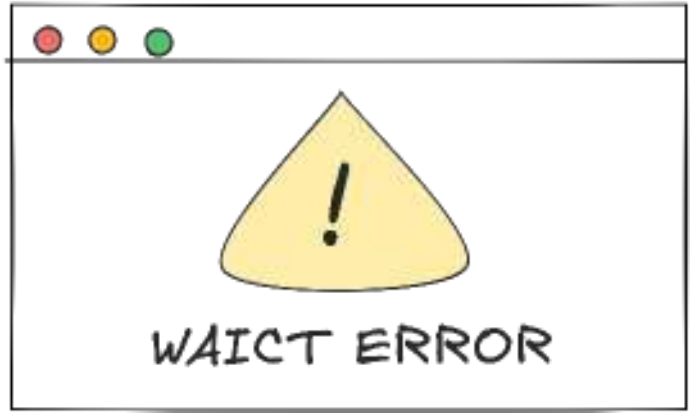

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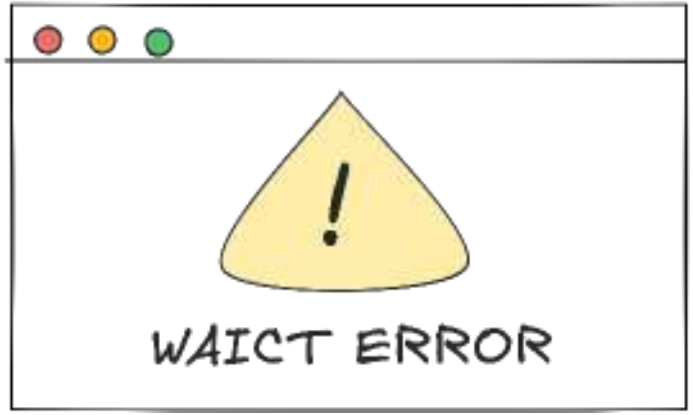

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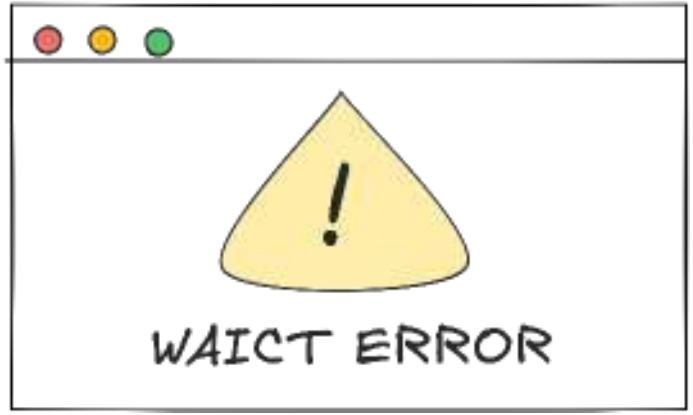

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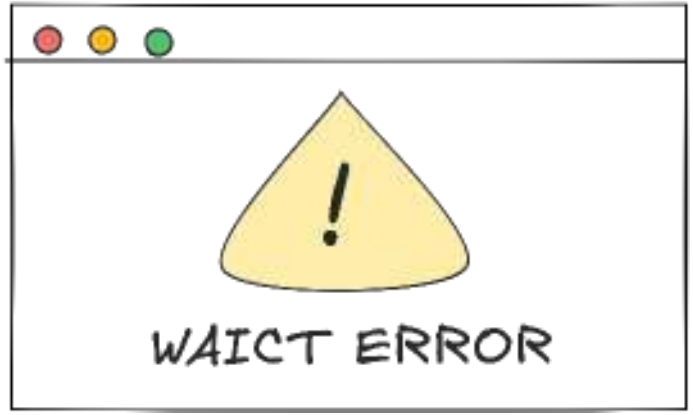

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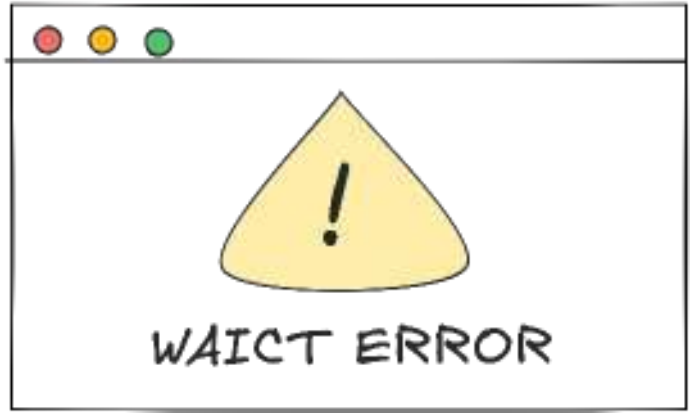

Immediate recovery through transparent opt-out

Running Code and Rough Specs

Running Code and Rough Specs

Firefox Nightly

Credits:
Tom Schuster
and Anna Weine

Running Code and Rough Specs

Firefox Nightly

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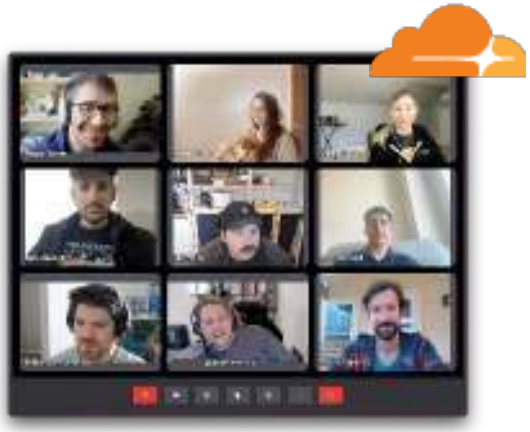


Orange Meets

Credits: Claude

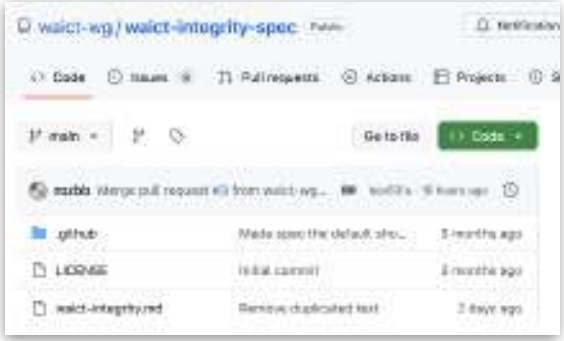
Running Code and Rough Specs

Firefox Nightly

Credits:
Tom Schuster
and Anna Weine

Orange Meets

Credits: Claude



Draft Specs

Trustworthy Javascript for the Web

CodeVerify

WEBCAT

WAICT

Ezzudin Alkotob

Giulio Berra

Demos, Specs and Slides

rwc26.waict.dev

Dennis Jackson

Michael Rosenberg

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